# NEEDLE: A Live Benchmark of Web Search APIs for AI Agents

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## Abstract

AI agents call web search APIs millions of times a day, yet no public benchmark tracks how well those APIs retrieve, and any fixed query set goes stale or gets tuned against. We present NEEDLE, an open-source benchmark that draws its queries and gold from public sources on every run (rare: a fixed stream), and a measurement study of the 26 search API configurations it has run: hourly news queries from RSS feeds and Google Trends, and daily known-item or fact queries from six public registries and archives. An LLM judge or a deterministic matcher scores each result, so no annotators are needed. A pooled "ceiling" normalizes every bench: engines are compared as a share of an oracle list pooled from all of them. On the 13 days when all 21 current engines ran, shares of the ceiling span 14–73%, weekly leaderboards are stable, body-detail queries separate engines far more than title queries, and a `site:` restriction on the same caselaw captions costs a third of recall. Blinded raters of three vendors agree with each other more than any agrees with the judge ($\kappa$ .90–.94 against .77–.79); every scorer is audited by LLMs, not humans. Paired tests separate most engine pairs, but the leaders form one neighbourhood, and on the one day scored after the rules were frozen Perplexity edges past Keenable, so readers should choose by group and by bench, not by rank. Code, data and dashboard: https://keenableai.github.io/needle.

## 1 Introduction

Agents rely on retrieval APIs that return a handful of URLs with titles and snippets; the choice shapes what the agent can know, yet developers pick on anecdote and price, because evidence is thin. Public IR test collections target document rankers, not live web indexes, and their labels age ("Fed rate decision" changes every six weeks). Agent benchmarks such as BrowseComp score the whole agent, so a weak retriever hides behind a strong model.

NEEDLE takes a different route (Figure 2). Its unit of evaluation is one API call: a query in, up to $k$ results out, as the agent sees them. Every run *generates* its queries from public sources, with a gold identity where one exists and a judged relevance profile otherwise, so no fixed test set accumulates beyond the rare stream and recurring registry facts (Limitations). Where gold has a canonical identity (an arXiv id, a reporter citation, a 10-Q figure), a deterministic matcher checks the returned URLs and snippets; where relevance is graded (news), an LLM judge rates each result and an ideal list pooled across engines normalizes the score. Scores rest on snippets and, for two benches, on one judge; we audit both (Limitations).

Contributions, each a claim the reader can check:

- A query recipe for five domains that yields fresh gold or judged relevance without annotators: three benches match gold identities deterministically (finance with an LLM backstop), and on the two judged benches two blinded sessions of one LLM (Claude Opus 5.5) that read a page excerpt agree with the judge at $\kappa$ .77 (§3.2, Appendix H).
- A scoring design that mixes a graded judge with exact matching and normalizes by a pooled ceiling; removing any one engine from the pool moves the overall share by at most .007 (§3.2).
- Four months of public results for 26 configurations: shares span 14–73% of the ceiling, and body-detail queries spread engines from .08 to .72 (§4, Figure 1, Table 1).

The combination is new; its scoring ingredients follow prior work (§2).

## 2 Related Work and Design Rationale

Each strand of prior work below fixes one design decision in NEEDLE.

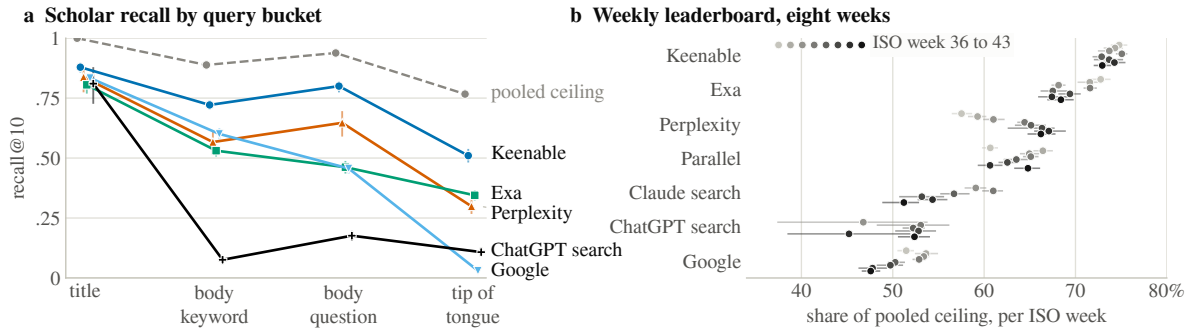



Figure 1: Engines that find a paper from its title diverge once the query names a detail from the body, and the weekly leaderboard holds its order while the pool grows. (a) Recall@10 on the four scholar buckets, 30 days to 29 October 2026 (31 runs, 332–420 queries each); dashed grey: pooled ceiling. (b) Overall share of ceiling per ISO week for seven configurations, one point per week from light (36) to dark (43); the pool grew from 15 to 21 configurations. Points: query-weighted means; lines: 95% run-level bootstrap intervals; raw scores.

**Fresh queries beat fixed test sets.** Static benchmarks leak into training data and get tuned against (Sainz et al., 2023; Balloccu et al., 2024). LiveBench, LiveCodeBench, RealTime QA, FreshQA and LiveSearchBench draw new items from dated sources or Wikidata deltas (White et al., 2025; Jain et al., 2025; Kasai et al., 2023; Vu et al., 2024; Zhou et al., 2025; Thakur et al., 2025); all score an answering model, or in FreshStack's case a retriever, over a fixed corpus. NEEDLE borrows the idea but studies the live retrievers.

**Agents need the retriever measured alone.** BrowseComp, GAIA and AssistantBench measure an agent end to end (Wei et al., 2025; Mialon et al., 2024; Yoran et al., 2024); BrowseComp-Plus freezes a corpus to separate retriever from agent (Chen et al., 2026), and the closest API study swaps three providers under one agent on a static set (Selvam and Ghosh, 2026). Agent query logs show quoted phrases and operators are common (Meng et al., 2026; Pezzuti et al., 2026), so half of NEEDLE's finance and legal queries carry operators, each engine receives them through its native filter where one exists (Appendix D), the news judge enforces them literally, and the known-item matchers accept the gold identity only on the requested host. Web search audits and overlap studies take one engine set at a few points in time, with fixed queries (Lewandowski, 2015; Bevendorff et al., 2024; Kirsten et al., 2026; Yagci et al., 2022); NEEDLE repeats a multi-API evaluation with regenerated tasks every run.

**LLM judges work at the system level, with known failure modes.** LLM assessors reproduce system rankings at TREC scale (Thomas et al., 2024; Upadhyay et al., 2024b,a; Arabzadeh and Clarke, 2025), so NEEDLE reports engine-level means only. Known risks shape the rest: circularity with LLM-based systems (Clarke and Dietz, 2024; Balog et al., 2025), keyword stuffing (Alaofi et al., 2024), a ceiling at the judge's competence (Soboroff, 2025; Faggioli et al., 2023), and snippet labels that diverge from full-text labels (Kazai et al., 2022); so identical results are judged once and snippets are capped uniformly.

**Known-item gold is free and synthetic queries are valid.** Simulated known-item queries date to Azzopardi et al. (2007), and LLM-elicited tip-of-tongue queries (Arguello et al., 2021) rank systems like real ones (He et al., 2025). Doc2query, InPars and Promptagator generate queries from documents (Nogueira et al., 2019; Bonifacio et al., 2022; Dai et al., 2023); NEEDLE does the same, with leak filters. Citation graphs give annotation-free gold for caselaw (Hou et al., 2025), and full text beats abstracts (Lin, 2009): the scholar buckets measure that effect live.

## 3 The Benchmark

### 3.1 Query generation

**News.** Each hour an LLM writes a terse query on the specific event of the newest item from each of 124 feeds and of the Google Trends RSS for 52 US geographies, and refuses evergreen items; each run then samples 20 queries round-robin over topical domains. **Finance.** Registry facts (CEO, founding year, headquarters, employees, LEI, ticker) come from Wikidata and GLEIF; quarterly revenue, net

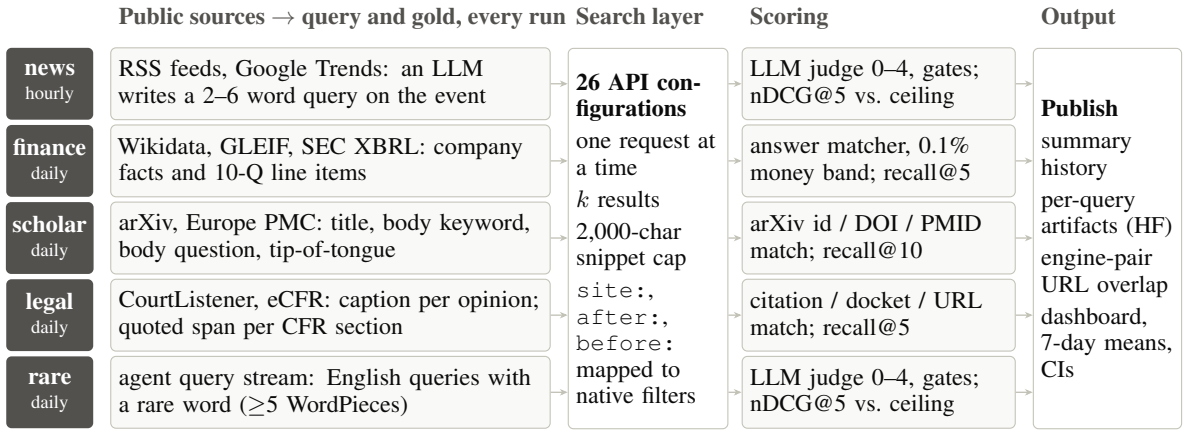



Figure 2: Every run regenerates its queries and gold (the rare bench samples a fixed stream and registry facts recur): fresh public items become queries with gold attached, go to every engine through one search layer, and are scored on the returned title, URL and snippet.

income, operating income and diluted EPS from SEC XBRL filings; a third set asks for a filing by description. **Scholar.** Each paper yields a *title* query with case and punctuation stripped; a *body keyword* query on a detail absent from title and abstract; a *body question* asking the same fact in prose; and a *tip-of-tongue* description with every name and exact value banned. A paper enters only if all four pass, so the buckets are paired. **Legal.** Recent opinions from 14 federal courts yield one caption-style query each; gold is the citation, docket number and CourtListener id. Sections of the Code of Federal Regulations (CFR) yield an LLM query built on one verbatim span of two to eight words, quoted in one variant in four; a query that leaks the citation is rejected (Appendix E). **Rare.** English agent queries from a public stream, with a word WordPiece splits into five or more pieces.

### 3.2 Scoring

**Graded relevance with a pooled ceiling.** For news and rare queries an LLM judge (GPT-6 at temperature 0) sees today's date, the query, a precomputed per-query profile (Appendix C) and one result (title, URL, date, capped snippet), and outputs a 0–4 rating adapted from the Needs Met scale of Google's rater guidelines (Google, 2025), with gates for integrity, entity, coverage, freshness and staleness, caps for sensitive domains, and literal compliance with quoted phrases and `site:`. Gains are $\{4:1, 3:0.667, 2:0.117, 1:0, 0:0\}$ (top-heavy, after ERR (Chapelle et al., 2009)) and repeated URLs score zero (Clarke et al., 2008). The *ceiling* is the best rating per deduplicated URL pooled across all engines, oracle-ordered. A judged query no engine answers with a rating of 2 or more is dropped; known-item benches keep every query (Appendix F).

**Identifier and answer matching.** Scholar and legal scan result URLs and snippets for arXiv ids, DOIs, PMIDs, reporter citations, docket numbers and CFR citations; recall@$k$ is the share of queries whose top-$k$ results carry the gold identity; for caselaw the identity is the case (opinion, docket page or a citing page), so the metric is case-identity recall (Appendix H). Finance asks whether the top-$k$ snippets *contain the answer*: nickname-tolerant names, a 0.1% band for money (table figures and parenthesized losses included) and 15% for headcounts, country aliases, domain match, and cue words for low-cardinality fields (year, country) so a stray number cannot score. An LLM backstop may upgrade rejected non-money results, never revoke a hit. The *ceiling* here is the pooled hit.

### 3.3 Aggregation and operation

An engine's headline number is $S(e) = \frac{1}{|B|}\sum_{b\in B} M_b(e)/M_b(\text{ceiling})$, its mean bench share of the pooled ceiling: a share of an oracle list, not of the web. Table 1 uses runs since 17 October where every configuration ran. A query issued in one run is a *query-run*.

**Reading Table 1.** The bench cells are absolute metrics (best per column in bold), the bar is the mean share of a pooled oracle, and everything is conditional on scorer version `paper-2026-10-30` (Appendix B): (i) news and rare are judged nDCG@5 on snippets; blinded raters of three vendors agree with the judge at $\kappa$

| Engine | Share of pooled ceiling | Scholar | News | Finance | Rare | Legal | Failed requests, % | | Latency |
|---|---|---|---|---|---|---|---|---|---|
| | %, mean and 95% CI | R@10 | nDCG@5 | R@5 | nDCG@5 | R@5 | service | account | $p_{50}$, ms |
| *ceiling (pooled)* | *100* | *.901* | *1* | *.993* | *1* | *.974* | | | |
| Keenable† | 73.0 | **.731** | .612 | .898 | **.504** | **.796** | 0.0 | – | 135 |
| Exa | 68.4 | .563 | .609 | **.915** | .472 | .771 | – | 1.9 | 2,098 |
| Perplexity | 66.6 | .578 | **.643** | .886 | .463 | .673 | – | 0.0 | 517 |
| Parallel | 65.1 | .587 | .553 | .901 | .385 | .739 | – | 0.8 | 2,588 |
| Brave (LLM ctx) | 64.2 | .533 | .640 | .847 | .449 | .658 | 0.0 | – | 1,114 |
| Linkup | 57.1 | .470 | .566 | .789 | .397 | .563 | 0.0 | – | 1,216 |
| Brave | 57.0 | .488 | .557 | .761 | .351 | .620 | 0.0 | – | 488 |
| Context | 56.8 | .598 | .476 | .778 | .326 | .577 | 0.0 | – | 891 |
| DeepSeek | 56.5 | .271 | .632 | .804 | .383 | .680 | 0.0 | – | 553 |
| You.com | 55.5 | .471 | .551 | .740 | .341 | .599 | 0.0 | – | 486 |
| Claude search‡§ | 50.9 | .499 | .464 | .664 | .318 | .525 | – | 35.0 | 34,323 |
| Firecrawl | 50.4 | .488 | .285 | .813 | .262 | .595 | 0.0 | – | 701 |
| Octen | 49.9 | .307 | .513 | .715 | .319 | .588 | 0.0 | – | 197 |
| Kagi§ | 49.6 | .560 | .318 | .717 | .299 | .508 | – | 11.9 | 1,275 |
| ChatGPT search‡ | 48.1 | .241 | .344 | .722 | .346 | .702 | 10.4 | – | 43,909 |
| Google (Serper) | 47.5 | .475 | .261 | .735 | .252 | .577 | 0.0 | – | 1,739 |
| Jina | 42.5 | .461 | .210 | .719 | .139 | .525 | 4.8 | – | 17,263 |
| Tavily | 42.2 | .303 | .303 | .821 | .277 | .359 | 0.1 | – | 1,953 |
| Bing (SearchAPI) | 42.1 | .416 | .292 | .630 | .170 | .533 | 0.3 | – | 2,008 |
| TinyFish | 37.5 | .122 | .255 | .728 | .213 | .523 | 27.2 | – | 1,899 |
| Ceramic | 13.9 | .001 | .040 | .427 | .111 | .111 | 0.1 | – | 98 |

Table 1: Shares of the ceiling span 14–73%; paired differences separate 187 of the 210 engine pairs, 179 after a nominal Bonferroni adjustment, 172 under every check (§4). 17–29 Oct 2026, the 305 runs in which all 21 active configurations took part (13.1k query-runs; Table 3). †The authors' engine. §Placement underidentified by account-side missingness: without the exclusion assumption Claude search spans 32.8–69.1 and Kagi 43.9–57.4 (Appendix H). ‡A model's server-side search tool; latency includes answer generation. Bar and whisker: mean bench share and its 95% run-level bootstrap CI. Bench cells: absolute metric, best per column in bold; italic row: the pooled ceiling. Failed requests: service-side failures score zero, account-side failures are excluded (Appendix H). Scorer version `paper-2026-10-30` (Appendix B; rules of Table 4); intervals are sampling uncertainty under this scorer, not scorer error. Latency: per-run median on news.

.77–.79 and with each other at .90–.94, and on 100 complete pools rank engines at Spearman .94 with the judge (Appendix H); (ii) scholar and legal are identifier matches; (iii) finance is answer presence under a 0.1% money band whose table reading audits at .70–.98 on a fresh window (Appendix G); (iv) a service failure scores zero and an account failure is excluded, which costs Claude search 35% of its query-runs; rescoring on the 6,801 queries no engine lost keeps the broad order ($\tau_b$ .91), and only Bing ($+5.7$), ChatGPT search ($-3.1$) and Parallel ($-1.9$) move beyond their intervals (Table 14); (v) intervals are run-level bootstraps; dropping the scored engine from the pool moves shares by at most .007.

## 4 Results

**The spread is large and bench-dependent.** Overall shares run from 14% to 73% of the ceiling. Paired run-resampled differences under the stated scorer (Appendix H) separate 187 of the 210 engine pairs pointwise; the 23 unseparated pairs form four neighbourhoods along the order. A Bonferroni adjustment leaves 179 separated (172 also under day resampling and matched support) and three neighbourhoods (Exa–Brave (LLM ctx), Linkup–You.com, Claude search–Bing) plus Bing–TinyFish; only Keenable and Ceramic stay separated from all others.

**Body-detail queries separate engines most.** Most engines find a paper from its title (Figure 1a, .71–.98); on body keywords Keenable keeps .72, Google .60, Exa .53, ChatGPT search .08.

**The leaderboard is stable but not static.** Adjacent weekly leaderboards over the eight weeks of Figure 1b agree at Kendall's $\tau$ .80–.93 and the leader did not change; on the one post-freeze day scored under the frozen rules Perplexity leads Keenable 72.4 to 71.3, within a day's noise (Appendix H).

**Four vendor pairs share most of their URLs.** The median pair shares .055 of its URLs (Jaccard; Figure 3, Appendix I); four pairs share .49–.64.

**Site restriction lowers caselaw recall.** The same 80 captions restricted to courtlistener.com lose .37 recall@5 (paired; 16 of 18 engines; Appendix J).

## Limitations

**Conflict of interest.** The authors build Keenable, which leads the current leaderboard; on the one day scored after the rules were frozen it is second to Perplexity (Appendix H). Everything that produces the numbers is public: the generator code, the judge prompts, the engine clients, every per-query result with the judge's reasoning, and the workflow that runs them; every table recomputes exactly from the archived ratings, while the judgments themselves can only be approximated (Appendix B). Keenable receives the same snippet cap, the same single-request pacing and the same judge as every other engine. The authors changed no engine's benchmark client during the window, Keenable's included; the paper's scoring rules were frozen on 30 October 2026, after the audits of Appendices G and H; every result in this paper is a rescoring of runs collected before that date. The public dashboard at the URL above still scores with the production rules of Table 4; the paper's rules will replace them as a labelled scoring version, and until then the two sets of numbers are not directly comparable. Whether any vendor's production index changed during the window, Keenable's among them, the benchmark cannot separate from ordinary development, and that holds for every engine alike. Readers should nonetheless treat the ranking as a claim to audit, and the artifacts exist so that they can.

**Snippet-only evidence.** The judge and the matchers see what the API returned, not the page. That view is what an agent sees before it fetches, and it keeps engines comparable, but it understates engines whose snippets are terse, and recall on the known-item benches is a lower bound on finding the identifier (Appendix H checks that hits are the target page). Snippet labels are known to diverge from full-text labels (Kazai et al., 2022), though judging what the consumer sees has its own case (Turpin et al., 2009) and LLM judgments from short text track full-text judgments well (Mohtadi et al., 2026). The pooled ceiling inherits the same bound. Snippet length does matter. On the runs of Appendix H, median snippets range from 116 to 2,000 characters; the six engines below 200 characters all sit in the lower half of the news ranking, yet the two longest sit last, and the rank correlation between median snippet length and news share is .33. Rejudging eight queries with every snippet cut to 200 characters moves engine shares by .08 on average and the order to $\tau_b = .73$ (Appendix H); the bench scores the snippet an agent actually receives, so we report this as a sensitivity, not a correction.

**The LLM judge.** News and rare scores depend on one judge model and one prompt. The prompt follows public rater guidelines and emits explicit gates and reasoning per judgment, and identical results are judged once so no engine is favored by sampling noise. A second judge from another vendor ranks engines at Spearman .90, and two blinded raters who also read the page agree with the judge at $\kappa$ .77 and rank the 21 engines at Spearman .89 on news but .71 on rare (query-resampled 95% intervals .74–.94 and .50–.94), while on 100 complete pools two further vendors rank them at .94 (.87–.97) with the judge, .91 on rare (Appendix H); the raters are LLMs of three vendors, not humans, they read the first 3,000 characters of a page rather than the document, with snippet and page in one prompt (a staged snippet-only pass on 400 items is in Appendix H), 6% of pages were not fetched, and they share the literal-quote rule with the judge, which lifts agreement on quoted rare queries ($\kappa$ .82) above the unquoted ones (.73), so this check bounds judge noise and, for the 1,130 fetched pages, snippet-only evidence, not shared model blind spots. LLM judges can be fooled by query terms in a snippet (Alaofi et al., 2024), favor systems that use a similar LLM (Clarke and Dietz, 2024; Panickssery et al., 2024), and cap measured quality at their own competence (Soboroff, 2025). The production judge shares a vendor with ChatGPT search and the second judge with Claude search; both engines sit in the lower half of the news column, and the second judge rated Claude search lower than the production judge did. Known-item benches do not use the judge and show the same overall leader, which is some reassurance, not proof.

**Finance matcher.** The matcher's earlier 2% money band and money backstop admitted neighbouring quarters and related line items: an LLM audit put their precision at .71 and .36, against .93 or higher elsewhere (Appendix G). Table 1 therefore uses a 0.1% band and no money backstop, rescored from the archived results; the change lowers finance recall by 2 to 11 points per engine and reorders the finance column ($\tau_b = .83$ against the old matcher) but leaves the overall ranking unchanged (Appendix H). An audit of 100 sampled misses of the tightened matcher found the gold value in 29, mostly unit-less table figures or parenthesized losses; reading those recovers 19 at .91 audited pre-

cision, and Table 1 includes that reading; on a fresh window (Appendix G) the reading's precision is .70 if a figure with an unlabeled period counts as wrong and .98 if only contradictions count; bounding the inflation per engine, or dropping the reading altogether, moves Firecrawl two ranks and no other engine more than two. Weighted by field prevalence and each engine's own money recall, the two errors do not cancel: reported finance recall runs 2 to 3 points high for the engines above .80, about 1 point high in the middle, and up to 1 point low for Claude search and Ceramic, which compresses the spread from .48 to .44 and swaps one adjacent pair (Appendix G); the audits pool engines, so per-engine matcher error is unmeasured.

**Coverage.** Queries are English and US-centric (US geographies for trends, US courts and regulations, SEC filers); other languages are unmeasured. The equal weight of the five benches in the headline share is a convention, not an estimate of agent traffic; Table 1 gives the per-bench columns for any other weighting, and Table 15 collects every alternative view of the order. The weighting matters for the lead: 3.7 of Keenable's 4.6-point lead over Exa comes from the scholar column (rare .6, legal .5, news .1, finance $-.4$); dropping any one bench keeps Keenable first ($\tau_b$ .86–.93 against Table 1) but shrinks the lead to 1.1 points without scholar (`bench_weights.py`), and within scholar the lead sits on papers under 30 days old and outside health sciences (Appendix H). On rare, the order also depends on the query mix: Exa and Perplexity lead the unquoted queries and Keenable the quoted ones (Appendix H). The news bench samples 20 queries per hour; single runs are noisy and should be read through the windowed means. Engines are called with default settings, a paid key and one configuration per vendor in Table 1; a vendor's best tier may differ. ChatGPT and Claude search are model-side tools that retrieve and then generate; we score the sources they return, so they are comparable on quality but not on latency or cost. Every query, gold value, result and judgment is archived with the judge model and engine list of its run, so any window can be re-scored offline from the stored ratings; re-running the judge is not exactly reproducible, because the published dates the APIs returned were not archived and runs are not stamped with their code revision, so the archive reproduces every table but only approximates the judgments themselves (Appendix H); the next release archives both.

**Gaming and repetition.** Query texts recur across runs. In the Table 1 window, counting every query-run whose exact text also appears in another run of the window, 11% of news query-runs recur (mostly within one day), 3% of legal, 5% of scholar, 17% of rare (sampled from a fixed filtered artifact) and 35% of finance (registry facts recur by design; `repeat_queries.py`); the last two are learnable in principle, and the finance facts are public registry entries in any case. The feeds are public, so a vendor could crawl them ahead of the hourly run. The LLM projection makes the exact query unpredictable, and a vendor that indexes the sources faster simply is fresher, which is what the bench measures; a vendor that detected benchmark traffic and answered it differently would not be caught, and snippets stuffed with query terms would raise judged scores (Alaofi et al., 2024); the page-text rater check (Appendix H) is the current guard against the latter. Per-query artifacts let readers check for either.

**Dispatch and support.** The comparison is live but not synchronized: each engine works through the query list in turn, so Claude search and ChatGPT search receive a news query 5 and 7 minutes after the fastest engine on average (Appendix B). On requests dispatched within 5 minutes of the first (96% of 100,839 engine–query pairs) the news order is identical and no engine moves by more than .01, so the lag is not measurable at this scale; a synchronized dispatcher is future work. Vendor response formats also change without notice: from about 02:00 UTC on 30 October 2026, the day after the window closed, Google (through Serper) and Jina returned `google.com/goto` redirect links instead of result URLs for about 40% of results (Appendix B), which URL-based matchers score as misses until the clients resolve the redirects. Account-side failures remove 35% of Claude search's query-runs and smaller shares for Exa, Kagi and Parallel; rescoring every engine on the same surviving queries over the whole window broadly preserves the order ($\tau_b$ .91) but moves Bing by $+5.7$, ChatGPT search by $-3.1$ and Parallel by $-1.9$ points (Appendix H), so the single ranks of those three engines depend on support. A reader of the dashboard sees ranks first; the support, failure and scorer notes of this paper belong beside them, and the dashboard will carry them.

**Ceiling as normalizer.** The ceiling is the best any evaluated engine found, not the best that exists. Adding a strong engine raises the ceiling and lowers everyone's share; shares are comparable within a window, not across engine-set changes. Unlike a TREC pool, every returned result is judged, so no engine loses credit for unjudged documents (Buckley et al., 2007); the pool only sets the normalizer. Rescoring each engine against a ceiling pooled without it (Webber and Park, 2009) moves no news share by more than .013 and leaves the order unchanged (Appendix H).

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## A Closest benchmarks

Table 2 sets NEEDLE beside the closest prior work on the dimensions a reader needs to place it: what one evaluation call is, how often the items refresh, where gold comes from, what is compared, and what is released.

## B Example queries and protocol

**Frozen protocol and manifest (scorer version `paper-2026-10-30`).** This paragraph is the authoritative specification of the paper's scorer; Table 4 dates the earlier production rules it replaces, and every table in the paper is computed under this version unless its caption says otherwise. The numbers were computed from the archived per-query artifacts at repository commit `c84faee3` with the rules stated here: a service-side failure scores zero and an account-side failure (quota, authentication or credit balance) is excluded, a `site:` hit counts only on the requested host, the finance matcher uses a 0.1% money band with table-figure and parenthesized-loss readings and without a money backstop, and the duplicate penalty applies to every query. The judge and the query generators are `openai/gpt-6-luna` through OpenRouter at temperature 0 with reasoning effort *high*; the second judge is Claude Opus 5.5. Snippets are capped at 2,000 characters before judging and matching. Scholar papers are balanced over five domains and three ages (7 days to 1 year). Table 3 and `paper/analysis/manifest.json` list, for every headline result, its data, runs, engine set and scoring rules. From the dataset revision named in the manifest (`6a5be09d`), two commands rebuild Table 1 with its intervals and the paired-comparison counts of §4:

```
python analysis/table_rows.py \
    history_v3.jsonl \
    table_results.tex
python analysis/adjacent_pairs.py \
    history_v3.jsonl \
    audit/adjacent_pairs.json
```

Table 1, Table 14 and Figure 1b come from `history_v3.jsonl` (the summary history rescored under these rules); Figure 1a from the scholar runs of the 30 days to 29 October; every audit in Appendices G and H names its runs and ships its sample and labels under `paper/analysis/audit/`. Two inputs of the original judgments are not recoverable: the published date each API returned for news results (not archived; its effect on ratings is measured in Appendix H) and the code revision of each run before 30 October 2026. Stored per result are the query, its profile, the title, URL and snippet, the judge model and the rating with its reasoning, so every table rescoring is exact and a judgment replay is approximate. The rare stream is the `agentic/queries.parquet` artifact of

| | Unit evaluated | Refresh | Gold | Systems compared | Released |
|---|---|---|---|---|---|
| FreshStack (Thakur et al., 2025) | retrieval over a fixed technical corpus | built once from recent questions | LLM-judged nuggets from community answers | retrievers and rerankers | corpus, queries, judgments |
| LiveSearchBench (Zhou et al., 2025) | answer to a question | periodic, from Wikidata deltas | entity facts from the knowledge graph | LLMs with search | questions, answers |
| API study (Selvam and Ghosh, 2026) | agent run with one search provider | once, static set | task answers | three search APIs under one agent | results tables |
| BrowseComp-Plus (Chen et al., 2026) | agent over a frozen corpus | once | human-verified answers | retrievers and agents | corpus, queries |
| NEEDLE (this paper) | one API call, $k$ results as returned | hourly and daily, every run | public identifiers and facts; LLM grades vs. pooled ideal | 26 live API configurations | code, prompts, per-query results and judgments, dashboard |

Table 2: NEEDLE beside the closest benchmarks on the dimensions that distinguish it.

the results dataset, built from the DeepResearch-Gym search logs (60% of sampled queries; Ning et al., 2026), the OpenResearcher trajectories (39%; Li et al., 2026) and LRAT-Train (1%; Zhou et al., 2026) and filtered as described in the repository (rare-word test, language identification, URL and identifier removal, near-duplicate cap). Engines are called through the clients of Table 8 at their default tier, one request at a time, with a 30-second timeout and up to three attempts with exponential back-off (1 to 30 seconds) on 429, 5xx and timeouts; a request that still fails is recorded with its error type and message.

**Dispatch order.** Within a run the engines are queried concurrently, and each engine works through the query list one request at a time, so fast engines finish the list before slow ones reach its middle. Across the 37 news and rare runs with per-query artifacts in the Table 1 window, the first and the last engine to receive the same query are a median of 16 minutes apart (90th percentile 54 minutes), a spread that the two slowest engines create while the other 19 stay within a few minutes of the first, and a whole run spans a median of 14 minutes (90th percentile 63); Ceramic is dispatched first most often, Claude search and ChatGPT search last. For hourly news the slowest engines therefore see the web up to an hour later than the fastest, an uncontrolled freshness advantage. To bound it we rescored the 249 news runs of the window that record dispatch times (4,959 queries, 100,839 engine–query pairs; `lag_subset.py`) on the pairs dispatched within 5 minutes of the first engine: 96% of pairs survive (ChatGPT search 1,958 of 4,959, Claude search 1,531 of 2,857), the order of the 21 engines is unchanged ($\tau_b$ 1.0), and no mean nDCG moves by more than .009 (Claude search $-.009$, query-resampled 95% interval $-.023$ to $+.003$; ChatGPT search $+.004$, $-.008$ to $+.016$). Restricting to the 3,728 queries whose dispatch spread is under 10 minutes, or to the 216 runs that finish within 15 minutes, also leaves the order unchanged with no mean moving by more than .005. Within an engine, the rank correlation between lag and nDCG has median .003 across engines. Mean lag is under 30 seconds for 18 engines, 2.3 minutes for Jina, 4.8 for Claude search and 6.6 for ChatGPT search. A synchronized dispatcher remains future work; at the present lags the effect is below the resolution of the window.

**Scoring rules by released result.** Table 4 lists the rules that differ between the public dashboard, which scores each run as it happens, and this paper, which rescores the archived per-query artifacts. The paper's rules are implemented in `paper/analysis/common.py`, and `paper/analysis/table_rows.py` is the exact entry point that produces Table 1 from the run manifest in the same folder; the dashboard history before 30 October 2026 uses the production rules.

**Chronology.** Table 5 dates every scoring decision, audit sample and new measurement against the evaluation window, so a reader can see which evidence was available when a rule was chosen and which came after.

**Release of agent-derived queries.** The rare stream is sampled from three published agent-trajectory datasets: the DeepResearch-Gym search logs of Ning et al. (2026) (`cx-cmu/deepresearchgym-agentic-search-logs`,

| Result | Runs | Scored units | Engines | Rules |
|---|---|---|---|---|
| Table 1, Table 14 | 305 (12 scholar, 258 news, 11 finance, 13 rare, 11 legal), 17–29 Oct | 13.1k query-runs | 21 | paper |
| Figure 1b | every run in ISO weeks 36–43 (pool grows from 15 to 21 configurations) | weekly query-weighted means | 7 shown | paper |
| Figure 1a | 31 scholar, 30 Sep–29 Oct | 332–420 queries per run | 6 shown | paper |
| Paired differences (§4) | Table 1 runs; 2,000 run- or day-level resamples | 13.1k query-runs | 21 | paper |
| Matched support | Table 1 runs (256 of 258 news) | 6,801 matched of 13,109 query-runs | 21 | paper, matched support |
| Dispatch lag | 249 news runs with timestamps | 4,959 queries, 100,839 pairs | 21 | paper |
| Blinded raters | 17–29 Oct runs with $\geq$19 engines | 1,200 results, 200 queries | 21, hidden | written guideline |
| Second judge | 8 news and 8 rare runs | 803 and 566 results | 23 | production prompt |
| List-level raters | 20 complete pools (10 news, 10 rare) from Table 1 runs | 1,150 results | 21, hidden | written guideline |
| List-level raters, two vendors | 100 complete pools (50 news, 50 rare) from Table 1 runs | 5,698 results | 21, hidden | written guideline, snippet only |
| Rare quoted vs. unquoted; bench weights; scholar strata | Table 1 runs | 1,300 rare queries; 5 benches; 4,820 scholar queries | 21 | paper |
| Caselaw identity | 10 legal runs, 18–28 Oct | 100 hits | 25 | LLM audit |
| Finance audits | 24 finance runs, 29 Sep–28 Oct | 399 results, 100 misses, 140 exact hits | 21 | LLM audit |

Table 3: Accounting for each headline result: the runs and scored units it uses, the engine set and the scoring rules. The machine-readable version is `paper/analysis/manifest.json`.

| Rule | Dashboard (production) | This paper |
|---|---|---|
| Duplicate penalty | skipped for `site:` queries | every query |
| `site:` hit | gold identity on any host | requested host only |
| Money band | 2% | 0.1% |
| Money readings | prose and plain figures | + table figures, losses |
| Money backstop | on | off |
| Account-side failure | scored zero | excluded |
| Service-side failure | scored zero | scored zero |

Table 4: Scoring rules by released result.

CC BY 4.0), the OpenResearcher trajectories of Li et al. (2026) (`OpenResearcher/OpenResearcher-Dataset`, MIT) and the LRAT-Train pairs of Zhou et al. (2026) (`Yuqi-Zhou/LRAT-Train`, no license stated on its card as of 30 October 2026). Their terms apply to the query texts; our own results data is CC BY 4.0. Attribution does not make a missing license a permission, so the next artifact revision drops the LRAT-derived queries unless the authors grant one; they are 113 of the 6,901 filtered queries and 19 of the 1,300 rare query-runs of Table 1, and dropping them changes no engine's mean rare nDCG by more than .009 (`rare_drop_lrat.py`). The filter keeps English queries with a rare word, removes URLs and identifiers and caps near duplicates. A regular-expression scan of the 1,988 rare queries released with the window's runs found no e-mail address, URL or key-like string; queries do name people, as the public source documents do. The repository issue tracker takes removal requests; a removed query leaves the current artifact and all future runs; the dataset history is squashed so that earlier revisions do not retain it, a public log records the query id and date of each removal without the text, and a paper snapshot drops the query as well, records the removal and recomputes its tables under a new snapshot version id listed in `manifest.json`, so a changed table is traceable to the removal. The artifact's dataset card at the time of writing carries only the MIT license of our code and does not name the sources; the sampler scripts stored beside the queries (`agentic/*_sampler.py`) record the exact upstream files, and the camera-ready card names the three sources with the licenses and citations above. The query texts stay under the upstream

| When (UTC) | Event | Status of the evidence |
|---|---|---|
| 17–29 Oct 2026 | Evaluation window of Table 1: every run in which all 21 configurations took part | archived per-query results and judgments |
| 26 Oct, 22:19 | Account-side failures excluded instead of scored zero (production and paper) | policy change inside the window; its effect bounded in Appendix H |
| 29–30 Oct | Finance matcher tightened to a 0.1% band without money backstop after the 80-hit and 50-upgrade audits; table-figure and parenthesized-loss reading added after the 100-miss audit | development samples; informed the rules |
| 30 Oct | 100 further exact-band hits and 59 other-field hits audited under the fixed rules | validation samples (.93, .98) |
| 30 Oct, 01:13 | Scorer version `paper-2026-10-30` frozen (score history generated); no rule changed after this point | every table in the paper uses this version |
| 30 Oct, 01:17 onward | Production runs continue online under the production rules; their archived outputs are rescored offline under the frozen paper rules (28 news, one scholar, rare, legal and finance run by 31 Oct 03:08) | prospective day (Appendix H) |
| 30 Oct, 02:56 | Google and Jina begin returning redirect URLs | vendor change; affects post-window runs only |
| 30 Oct, 08:49–22:53 | Paired operator runs on filing, caselaw and CFR targets | new measurement (Appendix J) |
| 30 Oct, 19:00 | Instrumented scholar generator run | new measurement (Table 16) |
| 30–31 Oct | Blinded raters: two Claude sessions (1,200 items), staged snippet-only pass (400), Gemini 4 and Grok 4.7 (1,200 items; 100 pools, 5,698 results) | validation of the judge, all after the freeze |
| 31 Oct, 00:47–01:44 | Fresh finance run (307 queries, 17 configurations) and audit of 489 hits | fresh-window validation of the frozen matcher (.93 exact, .70–.98 reading) |

Table 5: Every scoring decision was made before 01:13 UTC on 30 October 2026 and every validation sample after it; the paper window closed on 29 October. Development samples are those that informed a rule; validation and fresh-window samples were drawn after the rule they test was fixed.

terms, and our CC BY 4.0 covers only our results, judgments and metadata. The regex scan does not rule out personal or sensitive content that matches no pattern, so a human reads the queries of each new artifact revision before release. The per-query artifacts also redistribute the titles, URLs and snippets the APIs returned; they are published as evaluation records under the vendors' terms as we read them, and an engine's archived outputs are withdrawn on its request; a withdrawal is listed in `manifest.json` with the affected run and query ids, and any table recomputed without them carries a new snapshot version, so the published numbers stay traceable.

**Stewardship.** The benches run from the repository's main branch, so every change to a generator, prompt, client or scoring rule is a merged pull request with a date, and the commit log is the changelog of the benchmark. Each archived run records its judge model and engine list but not the code revision that produced it; the rules in Table 4 are therefore dated by the merge that introduced them, and the next release will stamp each run with its commit. A new engine joins through a pull request that adds a client and a key; it is scored from its first run, and because Table 1 keeps only runs on which every listed configuration ran, a newcomer shortens that window until it has a full one. No maintainer or auditor outside Keenable has yet been arranged; the archived artifacts and the rescoring entry point are what would make an external audit possible. Vendor responses change under the benchmark: from the hourly news run of 02:56 UTC on 30 October 2026 (17 of 100 results; 0 of 100 at 01:53; 36 to 41 per 100 in every later run that day; `goto_onset.py`) Google results fetched through Serper, and Jina's, which draw on Google, carry `https://www.google.com/goto?url=...` redirect links in place of the result URL. The title and snippet are intact, so the news judge is unaffected, but the URL-based scholar, legal and filing matchers and the `site:` host check score such results as misses, and the dashboard shows the drop for those two engines until the clients resolve the redirects; the window of Table 1 ends on 29 October and contains no such URL. A vendor who disputes a score can open an issue with the run and query ids; the per-result reasoning is public, and a rule change that follows is a pull request like any other. One day of the benchmark issues about 30,700 search requests over the 26 configurations (12,480 news, 10,920 scholar, 3,120

finance, 2,600 rare, 1,560 legal) and up to 75,000 result judgments before deduplication. Vendor search prices vary by plan, so we report no search cost. For the judge we did not meter tokens; a judgment sends about 450 tokens of query and result after a 3,500-token instruction prefix that the provider caches, and returns a rating with reasoning whose length we did not record. At the list price of GPT-6 Luna on 21 October 2026 ($0.10 per million input tokens, $0.01 cached, $0.50 output), 75,000 judgments cost about $6 of input a day plus $0.50 per million output tokens, so a day of judging falls between roughly $15 and $80 depending on reasoning length; this is an estimate from prompt sizes, not a billed figure. An external group needs the public code, one API key per engine it wants to compare and an OpenRouter key for the judge; a single bench on a single day is a few hundred requests. The dashboard shows each run's failed requests next to its scores; dated leaderboard views that freeze one scoring version are planned.

| Bench | Cadence | Queries/run | $k$ | Metric |
|---|---|---|---|---|
| news | hourly | 20 | 5 | nDCG@5 vs. ceiling |
| finance | daily | 120 | 5 | recall@5 |
| scholar | daily | 420 | 10 | recall@10 |
| legal | daily | 40 + 0–20 | 5 | recall@5 |
| rare | daily | 100 | 5 | nDCG@5 vs. ceiling |

Table 6: Per-bench protocol. Every engine receives the same $k$ it is scored on; snippets are capped at 2,000 characters; scholar draws 7 papers per domain and age cell, 4 queries each; legal adds one caption query per opinion the 14 courts published that day.

Table 7 lists queries generated in the last week of October 2026 with the gold the scorer holds for each. News and rare queries carry no gold: the judge profile fixes what a result must contain, and the pooled ceiling supplies the ideal list.

## C Judge profile and gates

The news and rare judge runs in two steps, both with the same model. A *profile* call sees the query and today's date and returns: query type (*broad*, *specific*, *list*); domain (health, legal, financial, academic, tech, news, commerce, jobs, other); whether the query carries operators; one to three MAIN aspects, phrased as what a document must contain and never as the answer; auxiliary aspects; and a content archetype (*ephemeral*, *persistent*, *evergreen*) that sets the staleness rule. The profile is stored with the query and pasted verbatim into every per-result call, so all engines are judged against one reading.

The *judgement* call sees the profile, today's date and one result (title, URL, published date if the API returned one, snippet capped at 2,000 characters) and emits a YAML block with eight gates and a rating. Gates: integrity (pass, limit, fail), relevance (entity, jurisdiction, topic), coverage of the MAIN aspects, completeness (list queries only), current-event freshness (dated events only), staleness by archetype, a domain cap (health, legal, financial, commerce and others carry hard caps), and operator compliance (quoted spans must appear verbatim in the snippet or title; `site:` must match the URL host). A failed relevance or operator gate forces 0; a failed coverage or completeness gate caps the rating at 2; staleness lowers the rating by one or two levels, or to 0 for a wrong year or entity status; a dated-event query whose result covers the entity but not the event is capped at 1. The rating maps to the Needs Met labels as 0,1 FailsM; 2 SM; 3 HM; 4 FullyM. The judge is told to score facts against the page's own text and authority, never against its own knowledge of the answer. The prompts are in the repository under `src/needle/shared/prompts`.

## D Operator mapping and frozen configuration

Table 8 fixes the configuration behind every engine in Table 1: the endpoint, the variant or model where the vendor exposes one, and the request parameters that differ from the vendor default. Every engine is asked for $k$ results ($k = 10$ on scholar, 5 elsewhere) with its default tier; a shorter list is scored as returned.

Every query is parsed for the operators `site:`, `after:` and `before:` (ISO dates). Table 9 shows what each engine receives. Google (Serper), Bing (SearchAPI) and Jina get the raw string because the underlying engines parse these operators themselves. Quoted phrases are always passed through in the query text.

## E Leak and quality filters

**Scholar.** A title query needs at least four words and a specific title (a digit, acronym, hyphenated term or five content tokens). A body keyword query needs at least three words and at least two content tokens absent from the title and abstract; a body

| Bench | Query | Gold or profile |
|---|---|---|
| news | Paramount Warner Bros takeover closes 2026 | MAIN: the expected or actual closing date of Paramount's takeover of Warner Bros. in 2026; archetype *ephemeral* |
| finance, registry | nvidia number of employees | 42,000, $\pm$15% |
| finance, 10-Q | "servicenow" q3 fiscal 2024 operating income | \$418M, $\pm$0.1% |
| finance, filing | Huntington Bancshares Ann B. Crane director vote 2026 after:2026-03-10 before:2026-06-08 | accession 0000049196-26-000038 |
| scholar, title | optimally selling heterogeneous objects by quantity | arXiv 2610.36244 |
| scholar, body keyword | gap rent regularity | same paper |
| scholar, body question | Which Go City pass for New York does the paper cite as a quantity-menu example, and what restriction applies to each selected attraction? | same paper |
| scholar, tip-of-tongue | Wasn't there a paper about a seller offering different goods at prices based on how many a buyer takes, letting them choose which ones, and showing this can bring the most revenue under certain value patterns? | same paper |
| legal, CFR | "modify or waive any" county committees authority | 7 CFR 1417.2 |
| legal, caselaw | citation insurance v broan-nutone first circuit after:2026-07-14 before:2026-10-12 | docket 25-1426 next to a party token; CourtListener id 10961606 |
| rare | "Krasnogvardeysky District" Russia | judged; rare word splits into 7 pieces |

Table 7: One generated query per bench and bucket, with the gold or judge profile it is scored against.

| Engine | Endpoint | Variant or non-default parameters |
|---|---|---|
| Keenable | api.keenable.ai | default; `keenable-realtime` is the realtime variant |
| Exa | api.exa.ai | `type=auto`; `exa-instant` and `exa-lite` are the instant and lite types |
| Google | google.serper.dev (Serper) | default |
| Bing | searchapi.io (SearchAPI) | engine `bing` |
| Brave | api.search.brave.com | web search; `brave-llmcontext` is the LLM-context endpoint |
| Parallel | api.parallel.ai | default processor; `parallel-turbo` is the turbo processor |
| Perplexity | api.perplexity.ai | search API; `perplexity-fast` is the fast mode |
| Tavily | api.tavily.com | default |
| Linkup | api.linkup.so | default |
| TinyFish | api.search.tinyfish.ai | default |
| Octen | api.octen.ai | default |
| Ceramic | api.ceramic.ai | default |
| You.com | ydc-index.io | default |
| Firecrawl | api.firecrawl.dev | search endpoint, default |
| Kagi | kagi.com/api/v1 | default |
| Jina | s.jina.ai | default |
| Context | api.context.dev/v1 | default |
| DeepSeek | api.deepseek.com | search tool endpoint (`/v1/tools/search`), default |
| Claude search | api.anthropic.com | `claude-sonnet-5` with the `web_search` tool |
| ChatGPT search | api.openai.com | `gpt-5.5` with `web_search`, `search_context_size=low` |

Table 8: Frozen configuration of the Table 1 engines. Operator handling per engine is in Table 9; the clients are in `src/needle/shared/search/`.

question needs eight words and the same novelty test. A tip-of-tongue query needs twelve words, no quotes, no decimal or five-digit number, at most two content tokens shared with the title, and no coined term (acronym, camel case, digit-bearing word) that appears in the title or abstract. A paper enters only if all four queries pass. **Legal.** A CFR query needs three words, no citation form (*CFR*, *U.S.C.*, §, "section $n$", "title $n$"), no part or section number of the gold section, and exactly one span of two to eight words that occurs verbatim in the section text; the quoted variant keeps the span in quotes, the others strip them. **News.** The projection prompt refuses evergreen items and returns a 2–6 word query on the specific event; query ids hash the text and the hour.

## F Worked example of the ceiling

The gains are a design choice, top-heavy so that one FullyM result outweighs one HM result by half and an SM result by an order of magnitude;

| Engine | `site:` | `after:`/`before:` |
|---|---|---|
| Brave, Brave LLM-context | in query text | freshness range |
| Keenable | in query text | published-date tokens |
| Exa | includeDomains | start/endPublishedDate |
| Tavily, Octen, TinyFish | include_domains | start/end dates |
| Linkup | includeDomains | from/toDate |
| Context | includeDomains | freshness (after only) |
| Perplexity | domain filter | date filters |
| Parallel | include_domains | after_date only |
| Kagi | lens sites | after/before filters |
| You.com | include_domains | freshness range |
| Firecrawl | in query text | cd_min/cd_max |
| DeepSeek | site: OR site: | appended to text |
| ChatGPT, Claude | allowed_domains | in prompt text |
| Ceramic | dropped | dropped |
| Google, Bing, Jina | raw string | raw string |

Table 9: How the search layer passes query operators to each API.

Appendix H shows the engine order under linear and binary alternatives. For engine $e$ and query $q$ with ratings $r_1..r_5$,

$$\mathrm{DCG}_e(q) = \sum_{i \le 5} \frac{g(r_i)}{\log_2(i+1)},$$

$$\mathrm{nDCG}_e(q) = \frac{\mathrm{DCG}_e(q)}{\mathrm{IDCG}(q)},$$

with gains $g(4) = 1$, $g(3) = 0.667$, $g(2) = 0.117$, $g(1) = g(0) = 0$, and $\mathrm{IDCG}(q)$ the DCG of the pooled ideal list. A bench score $M_b(e)$ is the mean over scored queries in the window (a service-side failure scores 0), the ceiling's score is the same mean over ideal lists (1 for news), and the headline share is

$$S(e) = \frac{1}{|B|} \sum_{b \in B} \frac{M_b(e)}{M_b(\mathrm{ceiling})}$$

over the benches $B$ the engine ran. Two engines answer one query. Engine A returns $u_1$ (rated 4), $u_2$ (2) and $u_1$ again under a tracking parameter (rated 3, but URL normalization strips the parameter, so the repeat scores 0). Engine B returns $u_3$ (3), $u_1$ (4) and $u_4$ (1). The *ceiling* keeps the best rating per normalized URL, $\{u_1 : 4, u_3 : 3, u_2 : 2, u_4 : 1\}$, in oracle order, so $\mathrm{IDCG} = 1 + 0.667/1.585 + 0.117/2 = 1.479$. Engine A scores $(1 + 0.117/1.585)/1.479 = 0.73$ and Engine B $(0.667 + 1/1.585)/1.479 = 0.88$. An engine that returns fewer than $k$ results is scored on the list it returned, against an ideal list that still takes up to $k$ from the pool. Had no engine returned a result rated 2 or higher, IDCG would be 0 and the query would be dropped from every engine's mean. Ties in the pool are irrelevant because only ratings enter the ideal. The paper applies the duplicate penalty to every query. The production code skips it when the query carries `site:`, a rule meant for one-host result lists that can in principle let a list exceed 1; we rescored the rare bench without the exception for Table 1, which changed 3 of 1,748 `site:` engine–query pairs, and no news query in the window carried `site:`.

## G Finance matcher check

The finance matcher originally accepted a money figure within 2% of gold, a headcount within 15%, and an LLM backstop upgrade on any field. We checked both bands on the 24 daily finance runs from 29 September to 28 October 2026 (a longer span than the 11 runs of Table 1, since the audit predates the window), 21 engines, 56,678 scored engine–query pairs, of which 13,140 ask for a quarterly money figure and 4,393 for a headcount.

**Tolerance sweep.** Table 10 rescores the stored results with other bands. The 2% money band buys 9 recall points over a 0.1% band and 18 over exact matching; the 15% headcount band buys 10 points over 1%. Engine rankings barely move: Kendall's $\tau_b$ against the default band stays at or above 0.94 for money within 0.1–1%. Counting individual matched results, 21.7% of money matches and 35.6% of headcount matches under the default bands fail at 0.1% and 1%.

**LLM backstop.** The backstop judged 12,919 engine–query pairs the matcher had missed and upgraded 3,187 (24.7%); 2,621 of the upgrades are money queries (49% of money misses), against 1.8% of identifier misses.

| Money band | 0 | .1% | .5% | 1% | **2%** | 5% | 10% |
|---|---|---|---|---|---|---|---|
| recall@5 | .363 | .447 | .498 | .518 | **.541** | .583 | .629 |
| $\tau_b$ vs. 2% | .96 | .98 | .95 | .94 | 1 | .91 | .89 |
| Headcount band | 0 | 1% | 5% | **15%** | 25% | 50% | |
| recall@5 | .794 | .811 | .863 | **.915** | .919 | .935 | |
| $\tau_b$ vs. 15% | .73 | .77 | .81 | 1 | .93 | .79 | |

Table 10: Deterministic recall@5 over all engine–query pairs and rank correlation of the 21 engines against the default band, by tolerance.

**Precision audit.** We sampled 399 results: 80 money hits that depend on the band (0.1–2%), 140 money hits within 0.1% (the *exact* stratum: 40 in a first sample and 100 more drawn from the Table 1 window), 40 banded and 10 exact headcount hits, 59 hits on the other field types in proportion to their frequency, and 70 backstop upgrades (50 money, 20 other). One LLM annotator per item (Claude Opus 5.5, in sessions of 50 to 100 items, without the matcher's verdict or the engine's name) judged whether the result states the gold value, with reporting rounding allowed and a value for a different period or line item counted as wrong. The first pass showed the annotator the first 700 characters of the snippet; every rejected matcher hit (58) was then re-judged on the full 2,000-character text the matcher reads, and 21 of the rejections reversed, almost all tables or later sentences that the 700-character view had cut off. Table 11 gives the final labels. Weighted by hit frequency, precision of the deterministic matcher is .97 overall, .97 on headcounts, .98 on the other fields and .88 on money, where banded hits reach only .71 (95% Wilson intervals: money within 0.1% .87–.96, banded .61–.80, headcount banded .80–.97, other fields .91–1.00): almost every remaining false positive is a neighbouring quarter or a related line item (basic or adjusted instead of diluted EPS, adjusted net income) that falls within 2%, or a table whose column headers the snippet omits. One rejected hit had a gold value of 0 from a faulty SEC fact; two of the 2,218 money and headcount gold values in the window are zero, and both count against the matcher here. The backstop is lenient on money (.36) and sound elsewhere (.95).

**Consequence.** Table 12 summarizes the revisions. Money queries are 23% of finance queries. The audit led to the matcher used in Table 1, rescored from the archived results: a 0.1% money band, no backstop on money fields, and (after the false-negative audit below) two extra readings of a money figure, a bare table number in thousands or millions and a parenthesized loss, accepted only when the result also carries the field's cue words and the number is not a year or a percentage. Every component is covered by an audit: money hits within 0.1% (.93 on 140 sampled hits, Wilson .87–.96), the two extra readings (73 of 80 sampled hits confirmed, .91, .83–.96; four of the seven failures were tables that stop before the queried row, two a different period), headcount hits (.93–1.00), the other fields (.98) and the backstop on non-money fields (.95), which now supplies under 2% of finance hits. A figure within 0.1% can still be a neighbouring quarter's identical value, so money is the matcher's weak spot. Appendix H gives the effect of each step on the leaderboard.

**False negatives.** We also sampled 100 money query-runs the strict matcher scored as misses, across engines and runs, and asked an LLM annotator (Claude Opus 5.5) whether any of the five results states the gold value for the right company, period and metric. It found the value in 29 (95% interval .21–.39; near misses, a different period or line item, in 2 more); 20 of the 29 were at rank 1 or 2. Nearly all are formatting the matcher does not read: bare table figures in millions or thousands without a unit word ("801" for $801M, "185,808" for $185.8M), losses in parentheses ("(458)", "$(3.98)"), and prose rounding beyond 0.1% ("$30 million" for $29.5M). Reading unit-less table figures and parenthesized losses recovers 19 of these 29 and matches 1 of the 71 others; Table 1 includes that reading. About a tenth of the remaining misses still hold the answer, mostly as prose rounded beyond 0.1%; against this, .07 of exact-band hits and .09 of extension hits are false positives. The error rates are stratum-specific, so we apply them by stratum (`finance_bias.py`): money queries are 23% of the window's finance queries, headcount 8% and the other fields 69%. For each engine we take its money recall $r_m$ (its reported recall minus the headcount and other-field recalls of the production matcher, which the rescoring did not change) and its extension share $x_m$ among money queries, and form expected false positives $.23\,[.07(r_m - x_m) + .09x_m] + .08 \cdot .07\,r_h + .69 \cdot .02\,r_o$ and expected false negatives $.23 \cdot .10\,(1 - r_m)$ (false negatives were audited for money only, so the other strata contribute none). The net correction runs from $-.030$ for Keenable

| Stratum | $n$ | precision |
|---|---|---|
| money, within 0.1% | 140 | .93 |
| money, banded (0.1–2%) | 80 | .71 |
| headcount, exact | 10 | 1 |
| headcount, banded (1–15%) | 40 | .93 |
| other fields (id, name, year, country) | 59 | .98 |
| backstop upgrade, money | 50 | .36 |
| backstop upgrade, other | 20 | .95 |

Table 11: Share of sampled matcher hits (329) and backstop upgrades (70) that the LLM annotator confirmed on the full text the matcher reads.

| Revision | Rule | Audited precision (sample) | Effect |
|---|---|---|---|
| Production (dashboard) | 2% money band; 15% headcount band; LLM backstop on every field | money banded hits .71, money backstop upgrades .36 (80 and 50 items; development sample) | the rules Table 1 replaces |
| Strict | 0.1% money band; no money backstop; non-money backstop kept | exact-band money hits .93 (140; 100 drawn after the rule was frozen), headcount .93 (40), other fields .98 (59); fresh window of 31 October: exact-band .93 (100) | finance recall $-2$ to $-11$ points per engine; finance order $\tau_b$ .83 vs production; overall shares move $\leq 1.9$ |
| Strict + readings (paper) | + unit-less table figures and parenthesized losses | .91 (73 of 80; development sample); fresh window of 31 October: .70 with an unlabeled period counted wrong, .98 counting only contradictions (238 yes, 95 unclear, 6 no of 339) | finance recall $+0$ to $+7$ points; three adjacent finance swaps; overall shares move $\leq 1.3$ |

Table 12: Finance matcher revisions, their audit status and their effect on the tables. Only the production rules ran live; the paper rescores the archived results under the last row.

and Parallel ($r_m$ .94 and .90) through $-.013$ for Brave and Context ($r_m$ about .5) to $+.006$ for Claude search and $+.011$ for Ceramic ($r_m$ .22 and .08). Correcting every engine compresses the finance spread from .476 to .435 and swaps one adjacent pair. The per-engine miss samples (0 to 3 recovered misses each) cannot show whether the error rates differ by engine. Of the audit samples, the first 80 banded money hits, the 100 misses and the extension recheck informed the rule changes and are development samples; the 100 further exact-band hits and the 59 other-field hits were drawn after the rules were frozen and give the validation estimates (.93 and .98); the extension reading's .91 is a development-sample figure.

**Fresh window under the frozen rules.** To validate the frozen matcher on data it had never seen, we generated finance queries with the production generator on 30 October 2026, kept all 146 money queries (net income, operating income, diluted EPS) and a stratified 161 of the other fields, and ran the 307 queries through 17 configurations between 00:47 and 01:44 UTC on 31 October (`raw_search_run.py`, `raw_to_report.py`, `fresh_finance.py`, `audit/fresh/`; Claude search was excluded because the account's credit had run out, and Google and Jina carried the redirect URLs of Appendix B, which does not touch the text-based money and name matches but masks the host). The run also exposed a defect in the production amount parser: its regular expression backtracks on long runs of spaces, so table-like snippets took minutes each and the scorer did not finish in four hours; the fresh results were scored after collapsing runs of spaces, which changes no match. Two Claude Opus 5.5 auditors with the matcher's full 2,000-character text (an earlier pass that showed 900 characters mislabeled many table hits and was discarded) graded every reading hit (339), 100 exact-band money hits and 50 other-field hits; each item is the first matching result in a query's top five, the result that sets that query's hit, so the precision below is the precision of query-level recall@5 decisions, not of arbitrary matched results. Exact-band precision is .93 (93 yes, 6 unclear, 1 no), as in the earlier validation. The table-figure and parenthesized-loss reading is weaker and engine-specific: 238 of 339 hits state the gold figure for the right period (.70, Wilson .65–

.75), 95 show the figure in a table whose period columns the snippet does not label, and 6 contradict it; we treat the 95 as unresolved, not as verified hits, so precision is .70 if an unresolved hit counts as wrong and .98 if only contradictions do. The unlabeled hits concentrate where snippets are table dumps: Firecrawl 13 of 43 confirmed, Tavily 12 of 24, Parallel 36 of 48, Brave (LLM ctx) 29 of 39, against Exa 43 of 45 and Perplexity 28 of 32. Of the 50 other-field hits 39 are confirmed; 4 of the 11 others are `website` hits matched on the URL host, which the auditor did not see, and 2 are headcounts inside the 15% band the rule allows, so the rule-consistent figure is .90. To bound the effect on Table 1, we count every reading hit of the window per engine (`reading_correction.py`), multiply by the engine's fresh unconfirmed share (its own where 20 or more hits were audited, .30 otherwise) and subtract from finance recall: Firecrawl loses 5.6 points, Tavily 2.7, DeepSeek 2.3, Parallel 2.1 and Brave (LLM ctx) 1.8, every other engine under 1; the finance order keeps $\tau_b$ .91 and the overall order .97, ranks 1 to 11 do not move, Firecrawl falls from 12 to 14 and Tavily and Bing swap. The reading therefore inflates finance recall for engines that return table dumps by up to 6 points under the strictest reading, which the paper's rules keep; a period-aware reading is future work. A strict score that drops every table reading is published alongside (`strict_vs_readings.py`, `audit/strict_vs_readings.json`): finance recall falls by 0 to 7.1 points (Firecrawl 7.1, Parallel 6.7, Exa and Tavily 5.7, Brave (LLM ctx) 5.2, ChatGPT search 4.7, DeepSeek and Perplexity 4.6, the rest under 2), the finance ceiling from .993 to .989, the finance order keeps $\tau_b$ .87 and the overall order .95; ranks 1 to 11 are unchanged, Firecrawl falls from 12 to 14, Kagi rises from 14 to 12, and ChatGPT search–Google and Tavily–Bing swap.

## H Ceiling and judge checks

The appendices use every active configuration, so names absent from Table 1, such as Parallel Turbo or Exa Instant, appear here.

**Leave-one-engine-out ceiling.** On every sixth hourly news run from 29 September to 29 October 2026 (111 runs, 1,909 queries with a non-empty ceiling) we recomputed each engine's nDCG@5 against a ceiling pooled from the other engines only. Because every returned result is judged, the ceiling can only fall when an engine leaves the pool, so its share can only rise. The largest change is +.013 (DeepSeek, which joined on 17 October and so has 578 queries); every other engine moves by .004 or less, and the engine order is identical (Kendall $\tau_b = 1$). An engine beats its leave-one-out ceiling on at most 1% of queries. Pool bias in the TREC sense (Buckley et al., 2007) does not arise, since no result is unjudged; the pool only fixes the normalizer, and the normalizer does not depend on any one contributor. On the Table 1 window the same check for every bench (`loo_all.py`) moves no scholar, finance or legal share by more than .007, no news share by more than .012 and no rare share by more than .034 (ChatGPT search, Exa), swaps one adjacent pair on news and two on rare, and moves the overall share by at most .007 with its order unchanged. When removing an engine empties a judged query's pool (its results were the only rated ones) the query is dropped for that engine: 13 of 5,083 news and 25 of 1,133 rare engine–query cases, at most 7 for one engine (Keenable on rare), and none on the known-item benches, where the pooled hit is defined for every query.

**Failure policy.** Table 1 scores a failure returned by the service as zero, which is what an agent experiences, and excludes failures caused by our own account (quota, authentication and credit-balance errors), which are all of the failures recorded for Perplexity, Exa, Kagi and Parallel and 99% of those for Claude search. Scoring only answered requests for every engine instead moves the engines with service-side failures: ChatGPT search (11% failed) from 48.1 to 51.8, rank 14 to 11; TinyFish (27%) from 37.5 to 44.9, rank 20 to 17; Jina (5%) from 42.5 to 44.6. Every other engine moves by at most 0.5 points, and the two tables agree at Kendall $\tau_b = .93$. Table 13 counts the excluded query-runs by engine and bench. Because the exclusions leave Claude search and Kagi scored on fewer queries than the other engines, we also computed a strict common-query leaderboard on the per-query artifacts we downloaded (scholar 3 runs, news 33, finance 12, rare 13, legal 10; 5,160 queries): a query counts only if all 21 configurations answered it, so a failure by any engine removes the query for every engine. It keeps 73 of 1,232 scholar, 360 of 660 news, 703 of 1,440 finance, 391 of 1,300 rare and 288 of 528 legal queries, and agrees with the paper's rule applied to the same runs at $\tau_b = .81$. The first-ranked engine and the bottom two are

unchanged; the largest moves are Bing (41.8 to 49.7, rank 19 to 14), ChatGPT search (49.2 to 53.1, rank 15 to 11) and DeepSeek (58.0 to 54.1, rank 7 to 10). The strict rule removes the penalty for service-side failures as well, which is why Bing and ChatGPT search rise; the scholar subset of 73 queries makes the strict shares coarse. A matched-support rule that drops a query for every engine only when an account-side failure occurred, and keeps service-side failures as zeros, applied to the whole Table 1 window (256 of its 258 news runs and every other run; `matched_pairs.py`), retains 2,774 of 4,820 scholar, 2,319 of 5,101 news, 720 of 1,320 finance, 626 of 1,300 rare and 362 of 568 legal queries, 6,801 of 13,109 in all, and agrees with the paper's rule at $\tau_b = .91$. Three differences have run-resampled 95% intervals that exclude zero: Bing $+5.7$ ($+2.4$ to $+9.9$, rank 19 to 15), ChatGPT search $-3.1$ ($-7.4$ to $-0.5$, 15 to 17) and Parallel $-1.9$ ($-3.0$ to $-0.9$, 4 to 5); every other engine moves by at most 1.2 points with an interval that covers zero, and the top three and bottom two are unchanged. On matched support the paired test of the next paragraph separates 8 of 20 adjacent pairs, 189 of 210 pairs pointwise and 175 with the Bonferroni adjustment, again with only Keenable and Ceramic separated from every other engine. Across the three estimates (run-level, day-level and matched support; `pairs_survive.py`) 172 of the 210 pairs stay separated under the adjustment in all of them; the 7 primary separations that fail one of the two checks are Context–ChatGPT search, DeepSeek–ChatGPT search, Firecrawl–Bing, Octen–Bing and ChatGPT search–TinyFish (matched support), Linkup–You.com (day level) and Octen–Google (matched support), so the support rule touches comparisons with Bing and ChatGPT search and little else. We keep the unequal-support estimate as the primary one because it uses all 13,109 query-runs rather than 6,801, and print the matched share beside it in Table 1.

The archived error records classify the failures: of 6,626 failed requests in the downloaded runs, 30% are HTTP 401–403 responses from quota lapses on our own accounts (all of Perplexity's, Exa's and Kagi's failures) and 18% are the Anthropic API refusing Claude search requests because our credit balance was exhausted (1,213 of Claude search's 1,220 failures; the other seven are its content filter); together these account-side failures are 48% of the total. Service-side failures are 9% 429 rate limits (ChatGPT search, Linkup, TinyFish), 2% 30-second timeouts (Bing, Context), nine 5xx responses and a remainder of 400-class errors spread over engines. The split is made per engine and period from the error records: a response counts as account-side only when it names our quota, authentication or credit balance (the 401–403 responses of Perplexity, Exa and Kagi and the Anthropic credit refusals); the 429 responses of ChatGPT search, Linkup and TinyFish arrived while our plans were within their limits and count as service-side, scored zero. Requests are retried up to three times with exponential backoff before a failure is recorded (Appendix B). The account-side failures cluster by date, not by position in the query list: Kagi's 1,562 fall on 27–29 October (1,180 on the 28th), Claude search's 4,607 on nine days peaking on 23–25 October, Exa's 256 on four days and Parallel's 105 almost all on 22 October, and each engine's failures split evenly between the first and second half of the list (`failure_profile.py`; the 6,530 request-level annotations with error type are in `audit/failure_profile.json`). The production dispatcher issues one request per engine at a time and the five benches run at different hours, so the window's 429 responses arrived at one in-flight request per account, unlike the paired and fresh runs of Appendices G and J, which sent two to four concurrent requests to some engines and provoked limits the production runs do not reach; the archived `queried_at` stamps give the per-request timing. Account-side failures are excluded from the scores of Perplexity, Exa, Kagi, Parallel and Claude search, while the Fail column of Table 1 still reports them. In particular, the collapse of Claude search in the week of 19 October on the dashboard was our credit lapse, not a change in the service. Exclusion assumes the engine would have scored on the failed requests as it did on the rest; two imputations bound the alternative (`exclusion_bounds.py`): scoring every excluded request zero or scoring it at the ceiling puts Claude search's overall share between 32.8 and 69.1 around the reported 50.9 (29–45% of its requests excluded per bench), Kagi between 43.9 and 57.4 (49.6), Exa between 63.5 and 70.6 (68.4), Parallel between 63.7 and 67.0 (65.1) and Perplexity between 65.2 and 68.1 (66.6). Without the assumption Claude search's rank is undetermined and Kagi's range spans ranks 9 to 15; Exa, Parallel and Perplexity keep their groups under either extreme,

| Engine | scholar | news | finance | rare | legal |
|---|---|---|---|---|---|
| Claude search | 1,626 / 5,208 | 2,102 / 5,831 | 480 / 1,440 | 473 / 1,300 | 167 / 624 |
| Kagi | 420 / 5,208 | 780 / 5,831 | 120 / 1,440 | 200 / 1,300 | 42 / 624 |
| Exa | 0 / 5,208 | 80 / 5,593 | 120 / 1,440 | 0 / 1,300 | 56 / 624 |
| Parallel | 0 / 5,208 | 100 / 5,831 | 0 / 1,440 | 5 / 1,300 | 0 / 624 |
| Perplexity | 0 / 5,208 | 0 / 5,831 | 0 / 1,440 | 0 / 1,300 | 0 / 624 |

Table 13: Account-side failures excluded from scoring, as excluded / total query-runs per engine and bench over every run from 17 to 29 October (12 finance runs, hence 1,440); Table 1 keeps the runs in which all 21 configurations ran (11 finance runs).

though Exa's upper bound narrows the Keenable–Exa margin from 4.6 to 2.4 points.

**Common support and uncertainty.** All numbers in this paper are recomputed from the archived per-query artifacts with the rules stated here. Engines joined the benchmark at different dates, so Table 1 restricts every bench to the runs since 17 October, when the last engine joined, in which all 21 configurations ran: every engine is scored on the same runs against a pool of the same engines (the query-runs lost to account-side failures are the only gaps: in the Table 1 runs, 0% of theirs for Perplexity, 0.8% for Parallel, 1.9% for Exa, 11.9% for Kagi and 35.0% for Claude search; Table 13 counts them over the 12 finance runs of the window; summary rows in which the production scorer had counted a whole-run account-side failure as zero, Kagi's news and rare runs of 27–28 October, are rescaled to exclude it, which moves Kagi from 46.2 to 49.6 and no other engine; the matched-support check under *Failure policy* above drops those queries for every engine). Averaging each engine instead over all runs of the 30 days to 29 October it took part in (31 scholar, 645 news, 23 finance, 23 rare and 30 legal runs; 31.6k query-runs) gives an order that agrees with Table 1 at $\tau_b = .88$ (`thirty_day.py`); the largest moves are Bing (42.1 to 45.8, rank 19 to 17), Kagi (49.6 to 53.2, rank 14 to 12), TinyFish (37.5 to 40.3) and Claude search (50.9 to 53.6); the top three and the bottom two are unchanged. Over the eight weeks of Figure 1b, the 15 configurations present in every week give adjacent-week $\tau_b$ of .81–.96 (mean .91), slightly higher than the full set. Pooling the news ceiling over those 15 engines only, on the sampled runs of weeks 40–43, raises their weekly shares by .003 to .011 and changes no weekly order ($\tau_b = 1$), so the growing pool does not drive the trends in Figure 1b. Resampling days instead of runs, so that an event's hourly runs move together, widens the bootstrap intervals by 6% on the 30-day table; in Table 1 15 of the 20 adjacent pairs have overlapping marginal intervals; the paired test below is the right reading of the order.

**Empty ceilings.** A judged query that no engine answers with a rating of 2 or more has an empty ceiling and is dropped: 69 of the 5,101 news queries in the window (1.4%) and 176 of the 1,300 rare queries (13.5%; 149 of the 551 quoted and 27 of the 749 unquoted). Scoring those queries as zero for every engine and for the ceiling instead lowers every absolute mean nDCG in proportion (by 1.4% on news and 13.5% on rare) but changes no share of ceiling by more than .003 and no order (`ceiling_drop.py`): the share is invariant because the zeros enter the engine mean and the ceiling mean alike, so the published scores condition on at least one engine finding useful evidence and the absolute nDCG columns of Table 1 should be read as conditional means.

**Paired differences.** Overlap of two marginal intervals is neither a test of equivalence nor transitive, so we also resampled runs once and recomputed every engine's share on the same resample (`adjacent_pairs.py`, 2,000 resamples, Table 1 runs). Runs are not fully independent units: counting every query-run whose exact text appears in another run of the window (first occurrences included), 35% of finance query-runs recur (registry facts), 17% of rare, 11% of news (mostly within one day), 5% of scholar and 3% of legal; counting only occurrences after the first, the shares are 22%, 9%, 6%, 3% and 2% (`repeat_queries.py`, `audit/repeat_queries.json`); the day-level resampling below treats a day's runs as one unit, so the dependence it does not model is the cross-day repetition of 35% of finance, 17% of rare and 4% of news query-runs. Resampling clusters of identical query text instead of runs (for scholar, the paper behind the four queries; 1,088 scholar, 4,783 news, 1,033 finance, 1,184 rare and 558 legal

| Engine | Share | 95% CI | Δ matched | 95% CI | Sch. | News | Fin. | Rare | Legal |
|---|---|---|---|---|---|---|---|---|---|
| Keenable | 73.0 | 72.1–73.7 | +0.3 | −0.4, +1.1 | .811 | .612 | .904 | .504 | .817 |
| Exa | 68.4 | 67.3–69.4 | −0.4 | −1.4, +0.5 | .625 | .609 | .921 | .472 | .792 |
| Perplexity | 66.6 | 65.1–68.1 | −0.5 | −2.0, +0.8 | .641 | .643 | .892 | .463 | .691 |
| Parallel | 65.1 | 64.0–66.2 | −1.9 | −3.0, −0.9 | .651 | .553 | .907 | .385 | .759 |
| Brave (LLM ctx) | 64.2 | 63.2–65.1 | +0.8 | −0.1, +1.7 | .591 | .640 | .853 | .449 | .676 |
| Linkup | 57.1 | 55.8–58.4 | +0.7 | −0.5, +1.9 | .521 | .566 | .794 | .397 | .579 |
| Brave | 57.0 | 56.0–58.0 | +0.4 | −0.6, +1.2 | .541 | .557 | .766 | .351 | .637 |
| Context | 56.8 | 55.2–58.3 | −1.2 | −2.8, +0.2 | .664 | .476 | .783 | .326 | .593 |
| DeepSeek | 56.5 | 55.4–57.5 | −0.1 | −1.1, +0.9 | .301 | .632 | .809 | .383 | .698 |
| You.com | 55.5 | 54.5–56.5 | +0.4 | −0.5, +1.2 | .523 | .551 | .745 | .341 | .615 |
| Claude search | 50.9 | 49.3–52.2 | +0.2 | −0.6, +1.3 | .554 | .464 | .669 | .318 | .539 |
| Firecrawl | 50.4 | 49.0–51.6 | −0.1 | −1.4, +1.1 | .541 | .285 | .818 | .262 | .611 |
| Octen | 49.9 | 48.7–51.1 | −0.4 | −1.6, +0.7 | .340 | .513 | .720 | .319 | .604 |
| Kagi | 49.6 | 48.7–50.6 | −0.0 | −0.8, +0.8 | .621 | .318 | .722 | .299 | .521 |
| ChatGPT search | 48.1 | 44.4–51.1 | −3.1 | −7.4, −0.5 | .267 | .344 | .727 | .346 | .722 |
| Google (Serper) | 47.5 | 46.6–48.4 | −0.1 | −1.0, +0.7 | .527 | .261 | .740 | .252 | .593 |
| Jina | 42.5 | 40.5–44.1 | −0.7 | −2.6, +0.6 | .511 | .210 | .724 | .139 | .539 |
| Tavily | 42.2 | 41.3–43.3 | −0.1 | −1.1, +0.8 | .336 | .303 | .827 | .277 | .369 |
| Bing (SearchAPI) | 42.1 | 38.3–45.6 | +5.7 | +2.4, +9.9 | .461 | .292 | .635 | .170 | .548 |
| TinyFish | 37.5 | 36.2–38.6 | −0.2 | −1.5, +1.0 | .136 | .255 | .733 | .213 | .537 |
| Ceramic | 13.9 | 13.2–14.6 | −0.4 | −1.2, +0.3 | .002 | .040 | .429 | .111 | .114 |

Table 14: Table 1 in numbers: overall share of ceiling (%) with its run-level bootstrap interval, the change in share under matched query support over the whole window (every engine rescored on the 6,801 queries no engine lost to an account-side failure; Appendix H) with its run-resampled interval, and share of ceiling per bench.

clusters; `cluster_pairs.py`) separates 194 of the 210 pairs pointwise and 186 after the Bonferroni adjustment, with Keenable–Exa at $+4.5$ ($+3.5$ to $+5.5$), more than the run-level 187 and 179 and the day-level 186 and 178, because run and day resampling carry day-to-day drift that query clusters do not; the paper reports the run-level counts, the adjusted count is nominal in that it applies a normal approximation to the bootstrap distribution of each difference, and all three counts remain conditional on the scorer, the support policy and the chosen unit. The paired 95% interval of the difference excludes zero for 8 of the 20 adjacent pairs and for 187 of the 210 pairs overall. The 23 unseparated pairs lie in four neighbourhoods along the order: Parallel and Brave (LLM ctx) ($+0.9$, interval $-0.3$ to $+2.1$); Linkup, Brave, Context, DeepSeek and You.com (differences under 1.3 points); Claude search, Firecrawl, Octen, Kagi, ChatGPT search and Google (under 2.8 points, with ChatGPT search's wide interval bridging the chain); and Jina, Tavily and Bing (under 0.3 points). The chains are not equivalence classes: Claude search is separated from Google, and Kagi from Jina. Keenable, Exa, Perplexity, TinyFish and Ceramic are separated from every other engine; Exa leads Perplexity by $+1.8$ ($+0.6$ to $+3.2$) and Perplexity leads Parallel by $+1.5$ ($+0.1$ to $+2.9$). These intervals are pointwise. With a Bonferroni adjustment for 210 comparisons (normal approximation on the bootstrap standard deviation, $z = 3.68$) 179 pairs stay separated. The 31 adjusted-unresolved pairs lie in three neighbourhoods: Exa to Brave (LLM ctx) (Exa–Perplexity $+1.8$, Perplexity–Parallel $+1.5$, Perplexity–Brave (LLM ctx) $+2.4$, Parallel–Brave (LLM ctx) $+0.9$), Linkup to You.com as before, and one chain from Claude search to Bing (ChatGPT search–Bing $+6.0$ and Google–Bing $+5.3$ are its widest links), plus Bing–TinyFish ($+4.7$); only Keenable and Ceramic are separated from every other engine (`audit/adjacent_pairs.json` lists every pair). Resampling days instead of runs, so that the hourly news runs of one day move together, separates 7 of the 20 adjacent pairs, 186 of 210 pointwise and 178 adjusted (`audit/adjacent_pairs_day.json`). Table 14 adds, beside the Table 1 share, the change in share under matched query support over the whole window, with its run-resampled interval. Under matched support Bing ($+5.7$) rises into the Claude search–Google neighbourhood and ChatGPT search ($-3.1$) falls below Google, while Claude search itself moves by $+0.2$; comparisons that involve Bing, ChatGPT search or Parallel therefore depend on the support rule, the others do not.

**Where the scholar lead comes from.** The scholar column carries most of the headline lead (Limitations), so we split it by the paper's age bucket, domain and query bucket

| Alternative view | What moves | Order vs. Table 1 | Leader and lead over Exa |
|---|---|---|---|
| Matched query support (6,801 queries) | Bing $+5.7$, ChatGPT search $-3.1$, Parallel $-1.9$; others within $\pm 1.2$ | $\tau_b$ .91; 175 of 210 pairs separated (adjusted) | Keenable, $+5.2$ |
| Day-level resampling | intervals widen | 178 of 210 pairs separated (adjusted) | Keenable, $+4.6$ |
| Query-cluster resampling (identical text; scholar: paper) | intervals narrow | 186 of 210 pairs separated (adjusted) | Keenable, $+4.5$ |
| Excluded requests scored zero or at the ceiling | Claude search 32.8–69.1, Kagi 43.9–57.4, Exa 63.5–70.6 | Claude search undetermined, Kagi ranks 9–15; Exa, Parallel, Perplexity keep their groups | Keenable, $+2.4$ to $+9.5$ |
| Ceiling pooled without the scored engine | overall share $\leq .007$; rare share $\leq .034$ | overall order unchanged | Keenable, $+4.6$ |
| One bench dropped (each of five) | lead from $+1.1$ (no scholar) to $+6.1$ (no finance) | $\tau_b$ .86–.93 | Keenable in all five |
| Rare column from unquoted queries only | rare: Exa, Perplexity, Brave (LLM ctx) lead; Keenable fourth | overall $\tau_b$ .97 | Keenable, $+3.2$ |
| Snippets cut to 200 characters (8 news queries) | news shares move .084 on average | news $\tau_b$ .73 | leader and bottom three unchanged |
| Finance error corrected by stratum | finance spread .48 to .44 | one adjacent finance swap | Keenable |
| Finance without table readings (strict matcher; Appendix G) | finance recall $-0$ to $-7.1$ (Firecrawl, Parallel, Exa, Tavily most) | finance $\tau_b$ .87; overall $\tau_b$ .95; ranks 1–11 unchanged, Firecrawl 12 to 14 | Keenable, $+5.4$ |
| Fresh finance window, 31 Oct 2026 (307 queries, 17 engines; finance column only) | finance shares move .05 on average | finance $\tau_b$ .75; same five engines lead | Exa .96, Keenable .95 on finance |
| Post-freeze day, all benches (30 Oct 2026; 19 engines) | one scholar and one legal run move the leaders | overall $\tau_b$ .84; same five lead, Ceramic last | Perplexity 72.4, Keenable 71.3 |
| Rater grades instead of the judge (100 complete pools, Gemini 4 and Grok 4.7; not a full overall score) | engine nDCG shares move .06–.07 on average | Spearman .94 ($\tau_b$ .83–.85) | same five engines lead and same three trail; Keenable–Exa unresolved under all three scorers |
| Empty-ceiling queries scored zero | absolute news and rare means fall 1.4% and 13.5% | shares move $\leq .003$; order unchanged | Keenable, $+4.6$ |

Table 15: Sensitivity panel: how the Table 1 order and its lead change under the alternative views analysed in this paper. Every retrospective view that recomputes the full overall score keeps Keenable first and Ceramic last, while the one prospective day puts Perplexity first by 1.1 points; the two sampled checks (200-character snippets, rater grades) preserve the top group and the bottom three but not the order within the top group. The views that move the middle of the table are matched query support (Bing, ChatGPT search, Parallel), the rare query mix and the snippet budget.

over the 12 Table 1 runs (4,820 queries; `scholar_strata.py`). Keenable's recall@10 exceeds Exa's by $+.38$ on papers 7 days old (.79 against .41), $+.10$ at 30 days and $+.03$ at one year; by $+.20$ to $+.30$ in computer, life, physical and social sciences but $-.10$ in health sciences (.35 against .45, where Parallel leads); and by $+.01$ on title queries (where Parallel leads), $+.17$ on body keywords and tip-of-tongue descriptions and $+.33$ on body questions. The lead is therefore a freshness and full-text lead on recent non-medical papers, not a uniform one. The production generator does not log how many candidate papers fail the requirement that all four queries pass, so we ran an instrumented copy of it once, on 30 October 2026 at 19:00 UTC, with the production code and query writer (GPT-6) over the 15 domain-by-age cells, sampling 30 candidates per cell where production samples 21 (`scholar_rejects.py`, `audit/scholar_rejects.json`). It sampled 614 papers from the feeds; 42 fell to the title filter (35 generic titles, 7 without a usable title or identifier) and 572 reached the query writer, of which 323 (56%) passed all four queries. The rejections are the model writing no body question (157), an unfetchable body (41), a body question that repeats title or abstract terms (37), a leaking tip-of-tongue description (8), a missing body key-

| Domain | 7 d | 30 d | 1 y | all |
|---|---|---|---|---|
| computer science | 21/35 | 34/39 | 30/47 | .70 |
| physical sciences | 16/26 | 23/37 | 17/45 | .52 |
| life sciences | 11/29 | 18/40 | 18/46 | .41 |
| social sciences | 13/28 | 16/35 | 18/44 | .44 |
| health sciences | 26/33 | 28/42 | 34/46 | .73 |
| all | .58 | .62 | .51 | .56 |

Table 16: Health and computer science papers pass the scholar generator's four-query requirement most often, and paper age matters less than domain. Accepted/candidate papers per domain-by-age cell in one instrumented generator run (30 October 2026); margins are acceptance rates.

word (4) and one model error. Acceptance varies by domain more than by age (Table 16): .73 in health sciences and .70 in computer science against .41 in life sciences and .44 in social sciences, and .58 at 7 days, .62 at 30 days and .51 at one year. The quotas fix the proportions: each cell takes its first seven accepted papers from 21 candidates (30 in the instrumented run), so in the 12 scholar runs of Table 1 every cell holds 6 to 8 papers (one run has 3 in social sciences at one year and one has 4 in life sciences at 7 days), except health sciences at 7 days, which Europe PMC fills in only 8 of the 12 runs (`scholar_strata.py` reads the cells from the archived gold). The filter therefore changes which papers a cell holds, not how many: in life and social sciences fewer than half of the candidates pass, so those cells hold the papers whose bodies yield a leak-free question and description, while health and computer science cells hold a less selected sample. Whether the rejected papers would rank engines differently is untested: they have no queries.

**Known-item hits are the target page.** A matcher hit shows that a returned result carries the gold identifier, not that it is the target document. On the three scholar runs of 25–28 October (13,311 hits) the identifier appears in the hit's own URL for 95.1% of hits, and 84% of hits are an arXiv page. On the ten legal runs, the section number appears in the URL of 94.0% of CFR hits and 95.4% sit on a primary legal host (eCFR, LII, GovInfo); for caselaw, 62.2% of hits carry the docket, cluster id or citation in the URL and 86.8% sit on a primary host (CourtListener, Justia, supremecourt.gov, uscourts.gov), while the rest are commentary pages that cite the docket (SCOTUSblog, Wikipedia, legal news). Because a primary host does not prove document identity, an LLM annotator (Claude Opus 5.5, engine names hidden) classified 100 caselaw hits drawn four per configuration from the ten runs: 80 are the opinion or a court docket page of the gold case on any host, 14 are pages about the same case (Wikipedia, news summaries, SCOTUSblog case pages, an oral-argument recording, a motion in the same docket), 5 are opinions in the same docket with a different filing date, and 1 is a page about another case. Per configuration, 12 of the 25 had the target page in all four sampled hits, 8 in three, 4 in two (Brave LLM context, Kagi, Linkup, Octen) and Ceramic in none; four hits per configuration bound rather than estimate per-engine identity rates. Known-item recall is therefore a tight measure of finding the document for scholar and CFR, and for caselaw it credits a page about the right case in roughly one hit in six. Scaling every engine's caselaw hits by the pooled target fraction .80 (Wilson .71–.87) lowers legal recall by .01 to .05 and keeps the legal order at $\tau_b$ .94 (`legal_target_only.py`); applying each configuration's own sampled fraction instead, which four hits bound only to within .25, gives $\tau_b$ .76, so the audit rules out a large pooled effect but not engine-specific ones. In the other direction, we sampled 100 scholar body and question query-runs the matcher scored as misses and had an LLM annotator check whether the target paper is among the ten results under a URL or snippet that carries no identifier: it is in 11 (95% interval .06–.19; project pages, aggregators such as ResearchGate, SciELO and PaperPeel, a podcast page and a news summary), never as a citation inside another document. Scholar recall is thus a lower bound by about a tenth of the misses, with no engine favored in the sample.

**Post-freeze day, all benches.** The production benches kept running after the rules were frozen on 30 October 2026, and their archived runs after the freeze give a one-day prospective window. The last rule change, the table-figure reading, was fixed at 01:13 UTC on 30 October 2026, when the paper's score history was generated; the first run after it is the 01:17 UTC news run, and the window runs to the finance run of 03:08 UTC on 31 October (28 news, one scholar, one rare, one legal and one finance run; 1,224 query-runs per engine), all dispatched after the freeze and rescored under the paper's rules (`prospective_window.py`, `audit/prospective_window.json`, `audit/prospective_history.jsonl`).

Nineteen configurations have all five benches: Claude search has no finance run because the account's credit had lapsed, and every Kagi request failed. The overall order agrees with Table 1 at $\tau_b$ .84 (per bench .73 to .87), the same five engines lead and Ceramic is last, but the leader changes: Perplexity 72.4 and Keenable 71.3, then Parallel 65.8, Exa 63.7 and Brave (LLM ctx) 61.7. The change comes from the single scholar run (Perplexity .64 to .77 share, Keenable .81 to .77) and the single legal run of 40 queries (Keenable .82 to .73, Perplexity .69 to .78), both within the run-to-run spread of those benches, while finance rises for every leader (.89 to .96) and news holds. Google and Jina returned redirect URLs for 51% and 50% of their results that day and fall 5 and 6 points. One day is a direction check, not a new leaderboard; the frozen rules will be applied to each further day as it is archived.

**Fresh window.** The 307-query finance run of 31 October 2026 (Appendix G), scored under the frozen rules, is a one-day prospective check of the finance column: its engine shares agree with the Table 1 finance shares at $\tau_b$ .75 with a mean absolute difference of .05 (`audit/fresh/summary.json`), and the same five engines lead (Exa .96, Keenable .95, Parallel .95, Brave (LLM ctx) .94, Perplexity .94, against .85–.92 in the table), now within .02 of each other. The sample is half money queries where the production mix is a fifth, it is one day, it has no LLM backstop for the non-money fields, and 17 of the 21 configurations ran, so it checks the direction of the finance column, not its values.

**Engine-specific scorer error.** Table 17 puts the two engine-specific matcher bounds developed above side by side. For finance it lists each engine's table-reading hits in the window, the share the fresh audit confirmed (the engine's own share where 20 or more hits were audited, the pooled .70 otherwise), the finance recall it would lose if every unconfirmed reading were wrong, and the resulting overall rank; for caselaw it lists the sampled fraction of hits that are the target case itself, legal recall under the pooled fraction and under the engine's own. The largest finance losses fall on Firecrawl, Tavily and DeepSeek outside the top five and on Parallel and Brave (LLM ctx) inside it, whose 2.1 and 1.8 points do not change their ranks; the engines whose caselaw hits are least often the target page (Ceramic, Octen, Kagi, Linkup) sit outside the top five. The two engine-specific errors therefore move middle and lower ranks by one or two places and leave the leading group and the bottom unchanged (`engine_error_table.py`).

**Sensitivity to scoring choices.** On the same 111 runs we recomputed news shares under other gain mappings and compared engine orders with the default by Kendall $\tau_b$: linear gains $\{1, .75, .5, .25, 0\}$ give .95 (one engine moves four places), the exponential mapping $(2^r - 1)/15$ gives .97, binary relevance at rating $\geq 3$ gives .93, at $= 4$ .87, and at $\geq 2$ .81. Treating queries with an empty ceiling as zero for every engine instead of dropping them gives .99 with no engine moving more than one place. Ranking Table 1 by the mean of the five absolute metrics instead of share of ceiling gives $\tau_b = .98$, swapping two adjacent pairs. For finance, Table 1 uses the matcher of Appendix G, rescored from the archived results of the window, and the path to it was taken in two audited steps. Tightening the earlier matcher (2% band with backstop) to a 0.1% band without a money backstop lowers finance recall by 2 to 11 points per engine (mean share of ceiling $-.056$) and reorders the finance column ($\tau_b = .83$; Exa leads under the old matcher because more of its money hits are banded or upgraded), while no engine changes rank in the overall table and no overall share moves by more than 1.9 points. Adding the table-figure and parenthesized-loss readings raises finance recall by 0 to 7 points per engine, moves overall shares by at most 1.3 points and swaps three adjacent pairs; with account-side failures excluded and `site:` hits confined to the requested host, Exa (.915) leads finance ahead of Keenable (.898). Deterministic tolerance alone moves no engine more than one finance place (Table 10).

**Blinded raters with page text.** The checks above reuse the production prompt. To test the judge against an independent reading, we sampled 100 news and 100 rare queries from the 17–29 October runs with at least 19 engines, drew six distinct top-5 URLs per query (1,200 results, judge ratings 0:441, 1:120, 2:263, 3:226, 4:150), fetched each page (1,130 fetched; the rest are login walls, blocked hosts or dead pages) and gave two Claude Opus 5.5 raters a written guideline (0–4 Needs Met scale with the same operator rules, in `analysis/audit/raters/`) and, per item, the date of the run, the query, the URL, the title, the snippet and the first 3,000 characters of the page, with the engine hidden, all in one prompt. Each

| | Finance: table-reading hits | | | | Caselaw: identity of the hit | | | |
|---|---|---|---|---|---|---|---|---|
| Engine | $n$ | confirmed | recall loss | rank | own share | any-host R@5 | × pooled | × own |
| Keenable | 57 | .79 | 0.9 | 1 | .75 | .80 | .76 | .75 |
| Exa | 117 | .96 | 0.4 | 2 | 1 | .78 | .73 | .78 |
| Perplexity | 98 | .88 | 0.9 | 3 | 1 | .69 | .64 | .69 |
| Parallel | 119 | .76 | 2.1 | 4 | 1 | .75 | .70 | .75 |
| Brave (LLM ctx) | 106 | .78 | 1.8 | 5 | .50 | .66 | .61 | .55 |
| Linkup | 9 | .70 | 0.2 | 6 | .50 | .56 | .52 | .46 |
| Brave | 5 | .70 | 0.1 | 7 | .75 | .62 | .58 | .57 |
| Context | 29 | .70 | 0.7 | 8 | 1 | .58 | .54 | .58 |
| DeepSeek | 100 | .70 | 2.3 | 9 | 1 | .69 | .64 | .69 |
| You.com | 5 | .70 | 0.1 | 10 | 1 | .60 | .56 | .60 |
| Claude search | 13 | .70 | 0.5 | 11 | .75 | .53 | .49 | .48 |
| Firecrawl | 111 | .33 | 5.6 | 12 → 14 | 1 | .59 | .56 | .59 |
| Octen | 34 | .78 | 0.6 | 13 → 12 | .50 | .59 | .56 | .52 |
| Kagi | 6 | .70 | 0.1 | 14 → 13 | .50 | .63 | .59 | .53 |
| ChatGPT search | 72 | .74 | 1.4 | 15 | 1 | .72 | .68 | .72 |
| Google (Serper) | 12 | .70 | 0.3 | 16 | .75 | .57 | .54 | .53 |
| Jina | 25 | .70 | 0.6 | 17 | .75 | .52 | .49 | .49 |
| Tavily | 84 | .58 | 2.7 | 18 → 19 | .75 | .36 | .33 | .32 |
| Bing (SearchAPI) | 3 | .70 | 0.1 | 19 → 18 | 1 | .52 | .49 | .52 |
| TinyFish | 18 | .70 | 0.4 | 20 | .75 | .52 | .49 | .48 |
| Ceramic | 2 | .70 | 0.0 | 21 | .00 | .14 | .13 | .09 |

Table 17: Engine-specific matcher error moves middle and lower ranks by at most two places. Finance: table-reading hits in the Table 1 window, the fresh-audit confirmed share, the finance recall lost (points) if every unconfirmed reading were wrong, and the overall rank before and after that loss (Appendix G). Caselaw: the sampled fraction of hits that are the target case (4 per configuration, so coarse) and any-host legal recall over the window's 11 legal runs (the diagnostic basis of Table 21, not the host-constrained Table 1 value) scaled by the pooled fraction and by the engine's own.

| Bucket | Syntax | $n$ | mean R@5 | ceiling |
|---|---|---|---|---|
| finance, money | plain | 332 | .777 | 1.00 |
| | quoted name | 335 | .756 | 1.00 |
| finance, filing | plain | 112 | .345 | .911 |
| | quoted phrase | 109 | .419 | .927 |
| | `site:sec.gov` | 113 | .649 | .991 |
| | `after:`/`before:` | 111 | .274 | .973 |
| legal, caselaw | plain | 64 | .885 | 1.00 |
| | quoted name | 24 | .823 | 1.00 |
| | `site:courtlistener.com` | 23 | .572 | 1.00 |
| | `after:`/`before:` | 17 | .689 | 1.00 |
| legal, CFR | plain | 200 | .478 | .945 |
| | quoted span | 100 | .609 | .990 |
| | `site:law.cornell.edu` | 100 | .661 | .990 |

Table 18: Recall@5 averaged over engines, and the share of queries any engine answered, by query syntax. Recall here is any-host and diagnostic; Table 1 counts a `site:` hit only on the requested host.

rater saw the 1,200 items in its own random order in 24 sessions of 50 and recorded a snippet grade and a page grade per item; the guideline asked for the snippet grade first, but the page text was visible while it was written, so the in-prompt snippet grade is not a staged measurement (the staged check is below). The 128 items with a missing page in Table 19 are the 70 pages no fetch returned plus 58 fetched pages whose text a rater marked as broken (cookie walls, boilerplate, error pages); 104 of the 128 are flagged by both raters. Table 19 gives the agreement. The two raters agree at quadratic $\kappa$ .94 (query-resampled 95% CI .93–.95). Their consensus agrees with the production judge at $\kappa$ .77 (.74–.81): .73 on news and .80 on rare, within one level on 86% and 91% of results. To test whether that gap is a Claude-specific reading, two raters from other vendors graded the same 1,200 items under the same guideline, one item per call at temperature 0 (Gemini 4 Argon and Grok 4.7 through OpenRouter; `third_rater.py`, `third_rater_analysis.py`,

labels_gemini.json, labels_grok.json). They agree with the Claude consensus at $\kappa$ .92 and .90 and with each other at .92, and each agrees with the production judge at .79 (query-resampled 95% CI .75–.82): .75 on news and .81 on rare, within one level on 89%. Raters of three vendors therefore agree with one another more than any agrees with the GPT-6 judge, and all three read the judge as stricter by about .3 of a level on news; at the engine level Gemini and Grok rank the 21 engines at Spearman .85 and .82 on news and .73 and .72 on rare against the judge, and at .98 and .98 overall against the Claude consensus. The judge is stricter: its mean rating is 1.60 against the raters' 1.97, and the gap is largest on news (1.83 against 2.32). Within the prompt the raters revised 7–8% of their own snippet grades after the page. A staged check (staged_page.py, audit/raters/staged/) re-graded 400 of the items with a fetched, unbroken page (200 news, 200 rare) in fresh Opus sessions that saw only the date, query, URL, title and snippet, under the same guideline with the page rules removed. That snippet-only grade differs from rater A's page grade on 25% of items (query-resampled 95% CI 21–29%; 28% on news, 22% on rare), three to four times the in-prompt figure, so the in-prompt snippet grades were pulled toward the page and the 7–8% understates the effect. The net shift stays small: page grade minus snippet-only grade is $+.05$ ($-.01$ to $+.11$; news $+.08$, rare $+.02$; 56 grades up, 43 down), so the page adds noise to a quarter of grades but no systematic bias at this sample size. The snippet-only grade agrees with the production judge, which also sees only snippets, at $\kappa$ .82 (.77–.85; news .78, rare .85), within one level on 92%, above the .78 of rater A's page grade on the same 400 items. At the engine level the judge's mean rating and the raters' consensus rank the 21 engines at Spearman .89 on news (query-resampled 95% interval .74–.94) and .71 on rare (.50–.94), with 37 to 61 items per engine, so the rare figure is coarse (rater_engine_ci.py). The absence of a net page effect is pooled over engines; the sample does not test engine-specific page bias. Because six results per query cannot give a list metric, a second rater pass graded 20 complete pools (10 news and 10 rare queries from the Table 1 runs, every top-5 list of all 21 engines, 1,150 unique results, snippet only; list_sample.py, list_analysis.py, audit/lists/). Recomputing each engine's pooled nDCG@5 from the rater grades and from the judge's ratings on the same lists ranks the 21 engines at Spearman .93 (query-resampled 95% interval .82–.97; Kendall $\tau_b$ .81) overall, .94 on news and .89 on rare (.67–.97), with a mean absolute share difference of .065; item-level agreement is .47 exact and .89 within one level, the raters again .19 of a level more generous. Twenty queries bound the headline differences coarsely, but the engines at the top (Exa, Perplexity, Brave (LLM ctx), Keenable) and bottom (Tavily, TinyFish, Ceramic) are the same under both scorers. The leading pairwise differences on these 20 pools keep their sign and overlap under both scorers (list_pair_diffs.py, query-resampled 95% intervals): Keenable–Ceramic $+44.5$ ($+28$ to $+62$) by the judge and $+50.7$ ($+37$ to $+65$) by the raters, Bing–TinyFish $+14.0$ and $+19.3$, TinyFish–Ceramic $+14.1$ and $+20.5$, Brave (LLM ctx)–Linkup $+7.7$ and $+6.7$; the near-top pairs are not resolved by 20 queries under either scorer (Keenable–Exa $-4.1$, $-16$ to $+8$, and $-8.0$, $-22$ to $+6$), so the scorer changes the strongest conclusions by less than their sampling width and leaves the close ones open. To tighten that bound we repeated the list-level check on 100 complete pools (50 news and 50 rare queries from the Table 1 runs, 5,698 unique results, snippet only) with two raters from other vendors, Gemini 4 Argon and Grok 4.7, one item per call under the snippet-only guideline (list_sample_big.py, api_grader.py, audit/lists100/). Recomputing every engine's pooled nDCG@5 from each rater's grades ranks the 21 engines against the judge at Spearman .94 (query-resampled 95% interval .89–.97; $\tau_b$ .85) for Gemini and .94 (.87–.96; $\tau_b$ .83) for Grok, .92 and .90 on news, .91 and .91 on rare (.77–.97 and .85–.97), with mean absolute share differences of .07 and .06; item-level agreement with the judge is .52 and .53 exact and .93 within one level, both raters about .3 of a level more generous. The same five engines lead under all three scorers (Keenable, Exa, Perplexity, DeepSeek, Brave (LLM ctx), in varying order) and the same three trail (TinyFish, Tavily or Kagi, Ceramic). The paired differences on these 100 pools keep their sign: Keenable–Parallel is $+20.4$ ($+15$ to $+27$) by the judge, $+16.8$ by Gemini and $+13.7$ by Grok; Keenable–Claude search $+18.6$, $+16.5$ and $+11.6$; Keenable–Ceramic $+53$, $+60$ and

+56; TinyFish–Ceramic +25, +35 and +32. Keenable–Exa is +3.5 (−2.4 to +9.0) by the judge, +3.7 (−1.6 to +8.9) by Gemini and 0.0 (−5.3 to +5.3) by Grok, so 100 pools still do not separate the top two under any scorer; Brave (LLM ctx)–Linkup and Bing–TinyFish, unresolved by the judge, are resolved in the same direction by both raters (+6.7 and +6.1; +3.9 and +6.7), the one place where the raters are more decisive than the judge. The rare sample exposes a generator defect: 39 of the 100 rare queries carry a double-quoted span, four of them wrapped as `query: "..."` by the source agent, and the literal-quote rule zeroes 64% of their results under the raters and 75% under the judge; judge–rater $\kappa$ is .82 on the quoted queries and .73 on the 61 unquoted ones (`rater_quoted.py`), so the shared rule lifts agreement where it applies. A wrapped query is scored on a phrase no page contains. The generator strips the wrapper from the first rare artifact released after 30 October 2026; runs before that date keep the queries as issued, and the four are flagged in the released rater sample. Over the whole Table 1 window (`rare_quoted.py`) 551 of the 1,300 rare queries carry a quoted span and 32 are wrapped; dropping the wrapped queries changes no rare share by more than .002. The quoted and unquoted halves rank engines differently: on the 722 scored unquoted queries (749 issued, 27 dropped for an empty ceiling) Exa, Perplexity and Brave (LLM ctx) lead and Keenable is fourth ($\tau_b$ .90 against the full rare order), while on the quoted queries Keenable, ChatGPT search and Exa lead and Parallel, Perplexity, Exa and Linkup lose .14 to .18 of their share ($\tau_b$ .60), so the rare column rewards literal compliance with quoted phrases as much as retrieval. The literal gate seldom punishes a page that does contain the phrase: of the 212 quoted-span results in the rater sample with page text, 74 show the span in the title or snippet and only 8 more show it on the page alone (`quoted_on_page.py`).

**Second judge.** We rendered the production prompts for every unique result of eight news queries, one from each of eight runs spread over the same month (803 results; 23 of the 25 configurations active then had complete lists for at least six queries), and had Claude Opus 5.5 rate them under the same instructions in six independent sessions, without access to the production ratings. We repeated the exercise on the rare bench: eight queries from the eight daily runs of 22–29 October, 566 unique results, 23 engines with complete lists for at least four queries. The artifacts do not store the published date the API returned, so the second judge saw title, URL and snippet only. To separate that input difference from the model difference we re-ran the production judge (same model, prompt and temperature) on the same rendered prompts without dates. On news it agrees with its own dated ratings exactly on 61% of results and within one level on 91% (quadratic $\kappa$ .80, query-resampled 95% interval .67–.88), and its mean rating falls from 2.43 to 2.19: the published date is a large input for news. Without dates it agrees with the second judge at $\kappa$ .76 (.68–.82), against .79 with dates, so the model difference is about as large as the date effect. On rare, where dates seldom matter, the rerun matches the dated ratings at $\kappa$ .97 and the second judge at $\kappa$ .91 (.81–.95). Table 20 summarizes agreement with the production judge (GPT-6). On news, ratings agree exactly on 55% of results and within one level on 92%; the quadratic-weighted $\kappa$ is .79 and the unweighted $\kappa$ .44. On rare, where half of all results are rated 0 by both judges, exact agreement is 73%, within-one-level agreement 99% and the quadratic $\kappa$ .90; engine shares correlate at Spearman .87 ($\tau_b$ .73) with a mean absolute difference of .061, and the second judge rates Exa, Linkup and Parallel Turbo .12–.14 higher. The second judge is stricter: its mean rating is 2.22 against 2.43, 68% of disagreements go that way, and 82% of all disagreements are by one level. Recomputing nDCG@5 for the eight queries under each judge, with the ceiling pooled from that judge's ratings, gives engine shares that correlate at Spearman .90 (Kendall $\tau_b$ .75; resampling queries gives a 95% interval of .76–.97 on news and .40–.95 on rare) with a mean absolute difference of .064; the two largest drops under the stricter judge are Parallel Turbo and Claude search (−.20 each), engines whose snippets often paraphrase the event without stating it. Eight queries are few, so these engine-level figures bound judge disagreement rather than estimate it precisely; as with human assessors (Voorhees, 2000), disagreement on items leaves the system ranking largely intact. Both judges read the same precomputed query profile, so a profile that misreads a query would not show up as disagreement. An annotator (Claude Opus 5.5) therefore wrote its own profile for the 16 sampled queries before seeing the production ones: 13 production profiles match in

| | News | Rare | All |
|---|---|---|---|
| results | 600 | 600 | 1,200 |
| page missing for a rater | 43 | 85 | 128 |
| $\kappa$, rater A vs. B (page) | .94 | .94 | .94 |
| $\kappa$, judge vs. consensus (page) | .73 | .80 | .77 |
| 95% CI | .67–.77 | .75–.84 | .74–.81 |
| within one level, judge vs. consensus | .86 | .91 | .88 |
| mean rating, judge / consensus | 1.83 / 2.32 | 1.38 / 1.61 | 1.60 / 1.97 |
| grade revised in-prompt after the page, A / B | .10 / .09 | .05 / .05 | .08 / .07 |
| staged snippet-only grade $\neq$ page grade (A), 400 items | .28 | .22 | .25 |
| $\kappa$, judge vs. staged snippet-only | .78 | .85 | .82 |
| Spearman $\rho$ on 21 engine means | .89 | .71 | .84 |
| $\kappa$, Gemini 4 vs. consensus / judge | .91 / .75 | .92 / .81 | .92 / .79 |
| $\kappa$, Grok 4.7 vs. consensus / judge | .90 / .75 | .88 / .81 | .90 / .79 |
| Spearman $\rho$ on 21 engines, Gemini / Grok vs. judge | .85 / .82 | .73 / .72 | .84 / .85 |

Table 19: Blinded raters of three vendors (two Claude Opus 5.5 sessions, Gemini 4, Grok 4.7) agree with each other more than any agrees with the production judge; a staged snippet-only pass shows the page changes a quarter of grades with no net shift, while in-prompt revisions understate the change. Quadratic-weighted $\kappa$; intervals from 1,000 query-level bootstrap resamples; consensus is the rounded mean of the two page grades.

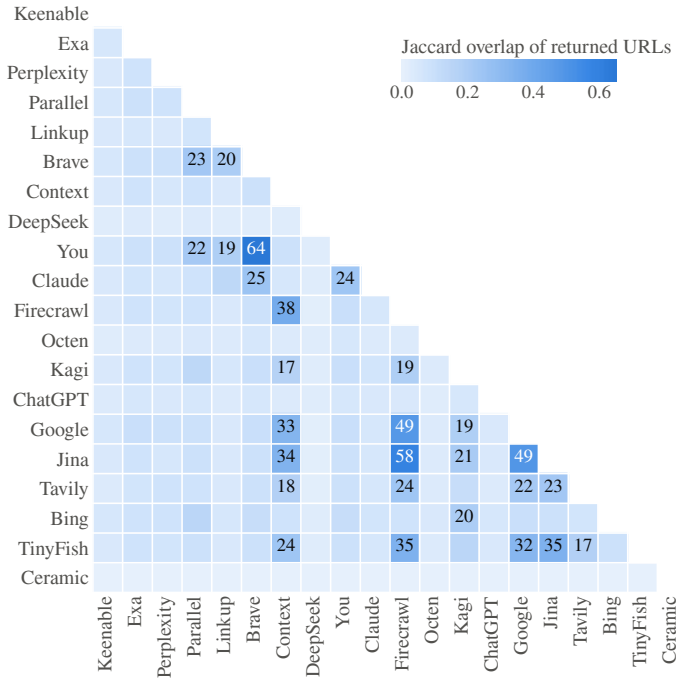



Figure 3: Four vendor pairs share about half of their URLs (Brave–You.com .64, Firecrawl–Jina .58, Google–Jina .49, Firecrawl–Google .49); the median pair shares .055. Jaccard overlap of returned URLs between the 20 vendors' base configurations over all runs, normalized per query and pooled, in Table 1 order; darker is more shared, and cells at or above .15 print their value in hundredths.

intent, two are narrower (they demand a detail the query only implies) and one is broader (it accepts any 2022 top-ten list of Arizona schools where the query names one school); none misreads the query or states the answer, and the query type differs for two. Profiles are stored with every query for inspection. The second judge's notes locate most disagreements in four gaps of the rubric: a title that answers the query over an empty or navigation-only snippet (integrity fail at 0 against a title-only cap at 2), reports written before the outcome a query asks about, how fast a deal from earlier in the year goes stale, and whether a rumor confirms an event. These are prompt fixes, not judge failures, and we will fold them into the rubric.

**Equal snippet budget.** To bound the effect of snippet length, the second judge rated the same 803 results again with every snippet cut to 200 characters. Against its own full-snippet ratings, exact agreement is .60, agreement within one level .94 and the quadratic $\kappa$ .80, with a mean rating of 2.14 against 2.22. At the engine level the cap compresses the field: engines with long snippets lose share (Exa $-.22$, Keenable $-.14$, Perplexity $-.13$) and engines with short snippets gain (Bing $+.14$, Context $+.10$, Google $+.09$); Kendall $\tau_b$ between the two engine orders is .73 and Spearman $\rho$ .88, with a mean absolute share difference of .084. Resampling the eight queries (`trunc_ci.py`) puts $\tau_b$ at .34–.83 and the mean absolute difference at .06–.13; the losses of Exa ($-.22$, interval $-.41$ to $-.08$), Keenable ($-.14$, $-.23$ to $-.05$) and Brave (LLM ctx) ($-.12$, $-.22$ to $-.04$) and the gains of Bing ($+.14$, $+.10$ to $+.19$) and Google ($+.09$, $+.04$ to $+.15$) exclude zero, while the leader's margin over the second engine does not under either cap ($+.01$, $-.12$ to $+.16$ with full snippets; $+.11$, $-.08$ to $+.32$ at 200 characters), so eight queries establish the direction of the length effect but not its size per engine. The leader and the bottom three do not change. Snippet length is therefore part of what the news bench measures, by design: an agent reads the snippet the API returns, and an engine that returns more useful text earns more. Readers who want a length-neutral view can rescore the public artifacts under any cap.

## I URL overlap

Figure 3 shows the Jaccard overlap of returned URLs between the 20 vendors' base configurations over all runs, computed on normalized URLs per query and pooled over queries. Four pairs exceed .45 and the median is .055.

## J Recall by query operator

The finance and legal generators assign syntax by cycling through a fixed list, so the variants of one bucket sample the same item population; they are not paired on the same item, so the first comparison below is unpaired. Table 18 gives mean recall@5 over engines and the pooled ceiling for each variant ($n$ counts scored query-runs), on the 24 finance runs of Appendix G and the 10 legal runs from 18 to 28 October 2026. The matcher does not require the returned host to match `site:`, so a gap measures lost recall, not disobedience. Unpaired, quoting the company name in a money query is nearly free on average (Octen loses 20 points), `site:` helps on filings and CFR sections and cuts mean caselaw recall by a third, and date operators cost 20 points on captions and 7 on filings; the paired run below tests the same contrasts on the same items.

**Paired variants (30 October 2026).** To pair the variants on the same item we took 80 filing-document, 80 caselaw and 80 CFR targets from the window's finance and legal query sets, stripped every operator and quotation mark to make a plain query, and wrote the quoted, `site:` and date variants of each (`paired_build.py`; the date variant brackets the filing or decision date by 45 days), then ran the variants of each target adjacently through 18 configurations in one run per bench on 30 October 2026 (`paired_analysis.py`, `audit/paired/`; no Kagi, You.com or DeepSeek). The run postdates the window; Google and Jina returned redirect URLs (Appendix B) on 43% and 36% of their results, so their rows in Table 22 measure that change too, and Keenable ran without an API key, so the public rate limit refused 91 of its 320 finance requests (229 answered). On the 80 filing targets the quoted span raises recall@5 by .10 on average (target-resampled 95% intervals exclude zero for 9 of 18 engines; none loses), `site:sec.gov` by .35 (16 of 18 gain), and the 90-day date window changes it by $-.02$, Linkup ($-.16$) and Brave (LLM ctx) ($-.15$) losing and Exa ($+.14$) gaining beyond their intervals, so the unpaired 7-point date cost does not survive pairing. On the 80 caselaw captions, `site:courtlistener.com` lowers recall@5 by .37 on average and for 16 of 18 engines beyond their intervals (ChatGPT search $-.88$, TinyFish $-.68$, Google $-.66$, Firecrawl $-.65$, Bing $-.62$, Context $-.61$, Exa $-.55$; Keenable $-.03$, Brave LLM ctx $-.03$, Brave $-.06$ and Perplexity $-.07$ lose least), so the unpaired third of Table 18 is a paired loss on the same items, not a difference between caption populations; the quoted caption changes recall by $-.04$ on average (Brave $-.16$, ChatGPT search $-.24$ and Keenable $-.09$ beyond their intervals) and the 90-day date window by $-.11$ (Google $-.35$, Brave and Linkup $-.33$, Tavily $-.25$, Brave LLM ctx $-.23$). On the 80 CFR sections the quoted span adds .18 (12 of 17 engines) and `site:ecfr.gov` adds .24 (10 of 17), the mirror image of caselaw (coverage, ranking and operator implementation are not separated here). In

| | News | Rare |
|---|---|---|
| results | 803 | 566 |
| exact | .554 | .730 |
| within one level | .918 | .995 |
| $\kappa$, quadratic weights | .790 | .898 |
| $\kappa$, unweighted | .440 | .584 |
| Spearman $\rho$ on ratings | .806 | .930 |
| $\kappa$, binary at $\geq 2$ / $\geq 3$ | .64 / .70 | .69 / .83 |
| engines | 23 | 23 |
| Spearman $\rho$ on shares | .903 | .874 |
| Kendall $\tau_b$ on shares | .746 | .728 |
| mean $\lvert\Delta$ share$\rvert$ | .064 | .061 |

Table 20: Agreement between the production judge (GPT-6) and a second judge (Claude Opus 5.5) on the same results and prompts, eight queries per bench.

| | Caselaw captions | | | CFR sections | | SEC filings | | | Host match on `site:` |
|---|---|---|---|---|---|---|---|---|---|
| Engine | plain | `site:` | dates | plain | `site:` | plain | `site:` | dates | finance / legal |
| Keenable | 1 | .91 | .94 | .64 | .90 | .39 | .70 | .36 | 1 / 1 |
| Perplexity | .98 | .87 | .94 | .53 | .75 | .55 | .79 | .44 | 1 / 1 |
| Parallel | .98 | .83 | .94 | .67 | .59 | .41 | .77 | .36 | 1 / 1 |
| Brave (LLM ctx) | .97 | .96 | .71 | .55 | .59 | .29 | .63 | .09 | 1 / 1 |
| Kagi | .97 | .91 | .53 | .59 | .58 | .36 | .34 | .17 | .19 / .09 |
| Exa | .96 | .55 | .93 | .64 | .84 | .63 | .81 | .56 | 1 / 1 |
| Brave | .94 | .96 | .59 | .51 | .61 | .29 | .60 | .15 | 1 / 1 |
| You.com | .94 | .96 | .59 | .47 | .62 | .23 | .61 | .15 | 1 / 1 |
| Linkup | .94 | .65 | .47 | .44 | .57 | .32 | .70 | .08 | 1 / 1 |
| DeepSeek | .94 | .70 | .82 | .54 | .76 | .35 | .39 | .39 | 1 / 1 |
| ChatGPT search | .92 | .26 | .94 | .61 | .79 | .51 | .78 | .60 | 1 / 1 |
| Claude search | .90 | .60 | 1 | .33 | .63 | .46 | .68 | .57 | 1 / .98 |
| Context | .88 | .26 | .65 | .34 | .77 | .29 | .80 | .21 | 1 / 1 |
| Firecrawl | .86 | .17 | .82 | .36 | .78 | .25 | .81 | .23 | 1 / 1 |
| TinyFish | .84 | .22 | .41 | .28 | .79 | .27 | .81 | .14 | 1 / 1 |
| Google (Serper) | .83 | .13 | .41 | .38 | .77 | .25 | .82 | .09 | 1 / .99 |
| Tavily | .83 | .22 | .47 | .21 | .44 | .21 | .65 | .07 | 1 / 1 |
| Jina | .80 | .29 | .12 | .33 | .80 | .30 | .80 | .14 | 1 / 1 |
| Octen | .77 | .17 | .47 | .57 | .60 | .10 | .30 | .15 | 1 / 1 |
| Bing (SearchAPI) | .72 | .19 | .71 | .51 | .52 | .30 | .56 | .17 | .95 / .86 |
| Ceramic | .23 | .30 | .06 | .10 | .11 | .01 | .02 | .01 | .06 / .03 |

Table 21: Recall@5 per engine on plain and operator variants of the known-item buckets (legal: 10 runs, 18–28 October 2026; finance filings: 24 runs), and the share of results on the requested host for `site:` queries. Recall here is any-host and diagnostic, not the host-constrained scorer of Table 1. Caption queries with date operators number 17, so those cells are coarse.

the legal run Claude search's caselaw cells rest on 30 targets and it has no CFR cell (credit lapse), Tavily's CFR cells on 29 and TinyFish's caselaw cells on 34 to 42 (rate limits), and Jina returned no results for 29 captions. Failures fall on whole targets (variants are dispatched together, in the order plain, quoted, `site:`, date), so each engine's difference uses the targets it answered; if those differ in operator effect from the rest, the estimate is biased in a way the interval does not show. Excluding Google and Jina leaves the caselaw `site:` loss at $-.36$ (14 of 16 engines beyond their intervals), and restricting to the 14 engines with all 80 captions answered gives $-.35$; the filing `site:` gain is $+.33$ and the CFR gains $+.14$ (quoted) and $+.20$ (`site:`) without Google and Jina.

**Operator compliance.** For every `site:` query we measured the share of returned URLs whose host is the requested one (Table 21). It is 1.00 for 21 engines on both benches, .95 and .86 for Bing, .19 and .09 for Kagi and .06 and .03 for Ceramic. The last two are limits of the vendor interface, not of the clients: Kagi's lens parameter (`sites_included`), its only host filter, does not restrict results, and Ceramic's API takes a bare query string, so the client strips the operators; passing them as raw text returns no results

| | SEC filings (80 targets) | | | | Caselaw captions (80 targets) | | | | CFR sections (80 targets) | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Engine | plain | quoted | site: | dates | plain | quoted | site: | dates | plain | quoted | site: |
| Perplexity | .66 | +.06 | +.07 | +.03 | .97 | −.03 | −.07 | 0 | .64 | +.14 | +.11 |
| Claude search | .54 | +.01 | +.20 | +.09 | .97 | −.10 | −.30 | 0 | – | – | – |
| Parallel | .53 | +.01 | +.29 | +.06 | .96 | 0 | −.14 | +.01 | .70 | +.09 | −.01 |
| Keenable | .52 | +.12 | +.28 | −.15 | .99 | −.09 | −.03 | −.03 | .64 | +.09 | +.25 |
| Exa | .51 | +.16 | +.28 | +.14 | .96 | −.03 | −.55 | 0 | .60 | +.14 | +.21 |
| ChatGPT search | .45 | +.05 | +.25 | +.05 | .95 | −.24 | −.88 | 0 | .56 | +.15 | +.16 |
| Brave (LLM ctx) | .29 | +.11 | +.38 | −.15 | .96 | 0 | −.03 | −.23 | .53 | +.05 | +.09 |
| Bing (SearchAPI) | .29 | +.01 | +.41 | +.01 | .91 | −.06 | −.62 | −.06 | .51 | +.11 | +.16 |
| Linkup | .26 | −.01 | +.54 | −.16 | .93 | −.01 | −.25 | −.33 | .45 | 0 | +.14 |
| Brave | .23 | +.10 | +.42 | −.06 | .97 | −.16 | −.06 | −.33 | .55 | 0 | +.06 |
| Firecrawl | .15 | +.30 | +.62 | +.04 | .89 | −.03 | −.65 | −.01 | .31 | +.39 | +.56 |
| Jina | .15 | +.20 | +.47 | −.06 | .61 | +.10 | −.18 | −.15 | .25 | +.44 | +.59 |
| TinyFish | .13 | +.19 | +.48 | −.04 | .88 | −.14 | −.68 | −.18 | .21 | +.47 | +.61 |
| Tavily | .12 | +.04 | +.39 | −.07 | .84 | +.04 | −.35 | −.25 | .55 | 0 | +.07 |
| Context | .11 | +.23 | +.44 | +.03 | .82 | +.04 | −.61 | −.03 | .30 | +.38 | +.55 |
| Google (Serper) | .07 | +.24 | +.56 | −.03 | .86 | −.10 | −.66 | −.35 | .26 | +.47 | +.53 |
| Octen | .07 | 0 | +.24 | 0 | .69 | +.04 | −.41 | −.06 | .56 | +.07 | +.09 |
| Ceramic | .00 | 0 | 0 | 0 | .19 | −.01 | −.19 | 0 | .14 | 0 | −.04 |

Table 22: A `site:` restriction helps on filings and CFR sections and hurts on caselaw captions; the quoted span helps on filings and CFR and costs a little on captions. Whether the caselaw loss reflects index coverage, ranking or the operator's implementation is not separated here. Plain recall@5 and the paired change under each variant on the same targets, one run per bench on 30 October 2026; a change printed in grey has a target-resampled 95% interval that includes zero, the others exclude it. Google and Jina returned redirect URLs on about 40% of results that day (Appendix B); Keenable's finance cells rest on 229 of 320 requests.

on all 20 test queries (`ceramic_rawops.py`). Google complies fully and still answers .13 of the `site:courtlistener.com` captions, so its gap is not operator handling. The stored results carry no published date, so date-operator compliance cannot be audited. Counting only `site:` hits on the requested host, as Table 1 does, lowers mean recall on `site:` queries from .578 to .527 on captions, .660 to .642 on CFR and .654 to .652 on filings; only Kagi (.91 to .00 on captions) and Ceramic change, every other engine by under .01. Table 21 reports the any-host figures.